

Year One Lecture Notes

Computer Science - Durham University

Matthew Emerson

May 2026

Contents

I	Algorithms and Data Structures	1
1	Fundamental Data Structures	1
1.1	Introduction	1
1.2	Basic Data Structures	2
2	Asymptotic Notation and Sorting	4
2.1	Asymptotic Notation	4
2.2	Sorting	5
3	Selection and Data Structures	6
3.1	Algorithms for Sorting, Searching, and Selection	6
3.2	Trees	6
3.3	Heaps	8
3.4	Lower Bounds for Sorting and Selection	10
4	Graph Theory and Algorithms	10
4.1	The Basics of Graph Theory	10
4.2	Paths, Cycles, and Connectivity	12
4.3	Trees and Isomorphism	13
4.4	Breadth and Depth First Searches	16
4.5	Minimum Spanning Trees	16
II	Computational Thinking	18
5	Introduction to Computer Science	18
6	Logic	18
III	Computer Systems	19
7	Digital Electronics	19
7.1	Circuits	19
7.2	Introduction to Logic	20
8	Machine Architecture	21
8.1	MIPS	21
8.2	The ATmega328P Microcontroller	22
8.3	Programming in AVR Assembly	22
9	Databases	22
9.1	Database Design	22
9.2	Modelling Databases	23
9.3	Relational Calculus and Algebra	24
10	Operating Systems	25
10.1	Introduction to Operating Systems	25
10.2	Process, Memory, and Storage Management	27
IV	Mathematics for Computer Science	31

11 Calculus	31
11.1 Introduction to Calculus	31
11.2 Multivariate Functions	32
11.3 Lagrange Multipliers	34
11.4 Series	34
12 Linear Algebra	38
12.1 Introduction to Linear Algebra	38
12.2 Matrices	40
12.3 Eigenvectors and Eigenvalues	41
12.4 Decomposition	45
V Discrete Mathematics (Elective)	50
13 Introduction to Discrete Mathematics	50
13.1 Arrangements and Permutations	50
13.2 Combinations	51
13.3 Arrangements and Combinations with Repetitions	51
14 Combinatorics	53
14.1 The Three Principles	53
14.2 Recurrence Relations	54
14.3 Generating Functions	57
15 Graph Theory	60
15.1 What is a Graph?	60
15.2 Eulerian Graphs	61
15.3 Planar Graphs	63
VI References	67
16 Algorithms and Data Structures	67
16.1 Module Professors	67
16.2 Cited Works	67
17 Computational Thinking	67
18 Computer Systems	67
18.1 Module Professors	67
18.2 Cited Works	68
19 Mathematics for Computer Systems	68
19.1 Module Professors	68
19.2 Cited Works	68
20 Discrete Mathematics	69
20.1 Module Professors	69
20.2 Cited Works	69

Part I

Algorithms and Data Structures

1 Fundamental Data Structures

1.1 Introduction

1.1.1 Basic Definitions

Definition 1.1.1.1: An algorithm is a method or process that is followed to solve a problem. It must be correct, must be composed of a finite number of unambiguous steps, and must terminate.

Definition 1.1.1.2: A data structure is a particular way of storing and organising data in a computer so that it can be used efficiently.

Definition 1.1.1.3: The RAM Model of Computation is a highly simplified model of computation that we use to compute the efficiency of an algorithm. Its memory consists of an infinite array which it uses to carry out instructions sequentially. All instructions take unit time, meaning the running time is the number of instructions executed.

Definition 1.1.1.4: Pseudocode is a generic way of describing how an algorithm will function without choosing any specific programming language. It does not have any very strict rules, but follows the following generic format:

Algorithm 1 Example pseudocode

Input: Input conditions

Output: The algorithm's output

```
 $x \leftarrow 1$ 
 $y \leftarrow 2$ 
if  $y > x$  then
    return  $y$ 
else
    return  $x$ 
end if
```

Definition 1.1.1.5: A variable is used to store some value, for example, in the above code, x and y are variables. Some basic types include integer, float, char, and string.

1.1.2 Recursion and Backtracking

Definition 1.1.2.1: A recursive algorithm is an algorithm which calls itself to do a part of its work. It should have a base case (where the final result is returned), must change its state to move towards the base case, and must call itself recursively.

Definition 1.1.2.2: A backtracking algorithm tries to construct a solution to a computational problem incrementally, one small piece at a time. Whenever the algorithm needs to decide between multiple alternatives to the next component of the solution, it recursively evaluates every alternative and then chooses the best one. Definition from (Erickson, 2019, p.71).

Definition 1.1.2.3: Memoisation is the process of storing the result of a computation so that it can be subsequently retrieved without repeating the computation.

1.2 Basic Data Structures

1.2.1 Arrays

Definition 1.2.1.1: An array is a sequence of elements that is usually denoted as $A[1]$, $A[2]$, ..., $A[n]$ or as $A[1, \dots, n]$. The elements are stored in consecutive memory cells and are all of the same data type.

1.2.2 Lists

Lists can be separated into three different categories, depending on how they store data.

Definition 1.2.2.1: A singly-linked list consists of nodes, each storing an element and a pointer to the subsequent node in the list. This means that the nodes do not have to be stored consecutively.

- **Definition 1.2.2.2:** The first node in a singly-linked list is called the head.
- **Definition 1.2.2.3:** The final node in a singly-linked list is called the tail and its pointer points to null.

Definition 1.2.2.4: A doubly-linked list consists of nodes with a value and two pointers, pointing to both the previous and subsequent nodes in the list. The list also must keep a size counter to track the number of nodes in the list.

- **Definition 1.2.2.5:** At either end of the doubly-linked list is a sentinel node. These nodes either have a null next or previous pointer and do not store a value. An empty list has two sentinel nodes pointing at each other.

Definition 1.2.2.6: A circularly-linked list uses the same nodes as a singly-linked list but does not have a head or tail node. Instead, the final node points to the start of the list, creating a circle.

- **Definition 1.2.2.7:** One node in the circularly-linked list is designated as the cursor and is used as a starting point when traversing the list.

1.2.3 Stacks

Definition 1.2.3.1: A stack is a collection of objects that are inserted and removed according to the last-in-first-out (LIFO) principle. They support the following methods:

- **Compulsory Methods:**
 - **Push:** $push(element)$ inserts $element$ at the top of the stack.
 - **Pop:** $pop()$ removes and returns the top element from the stack. An error occurs if the stack is empty.
- **Optional Methods:**
 - **Size:** $size()$ returns the number of elements in the stack.
 - **Is Empty:** $isEmpty()$ returns true if the stack is empty, and false if not.
 - **Top:** $top()$ returns the top item from the stack without removing it. Throws an error if the stack is empty.

Stacks are generally implemented using an array to store the items of the stack; however this can lead to memory wastage when the stack is not full, and conversely it will throw errors if the stack is full.

1.2.4 Queues

Definition 1.2.4.1: A queue is a collection of objects that are inserted and removed according to the first-in-first-out (FIFO) principle. They support the following methods:

- **Compulsory Methods:**

- **Enqueue:** *enqueue(element)* inserts *element* at the end of the queue.
- **Dequeue:** *dequeue()* removes the element from the front of the queue. An error occurs if the queue is empty.

- **Optional Methods:**

- **Size:** *size()* returns the number of elements in the queue.
- **Is Empty:** *isEmpty()* returns true if the queue is empty, and false if not.
- **Front:** *front()* returns the element from the front of the queue without removing it. Throws an error if the queue is empty.

1.2.5 Hash Tables

Definition 1.2.5.1: A hash table consists of a bucket array and a hash function.

Definition 1.2.5.2: A bucket array for a hash table is an array A of size N , where each cell of A is thought of as a bucket storing a collection of key-value pairs.

Definition 1.2.5.3: The size N of the array is called the capacity of the hash table.

Definition 1.2.5.4: A hash function h is a function mapping each key k to an integer in the range $[0, N - 1]$, where N is the capacity of the hash table. This provides an index $h(k)$ of a location in A where the key-value pair (k, v) is stored. Hash functions are usually composed of the hash code (mapping keys to integers) and a compression function (integers to indices).

Definition 1.2.5.5: A collision occurs when two keys map to the same hash value, and consequently want to be stored in the same bucket. These can be dealt with using different methods:

- **Definition 1.2.5.6:** Separate chaining converts each bucket to a list of key-value pairs. This will result in lists of average length $\frac{N}{M}$ where N is the amount of data, and M is the capacity of the hash table.
- **Definition 1.2.5.7:** If a collision occurs at $h(k) = i$, linear probing tries to insert again at $A[(i+1)\%N]$, then if that is also full, tries at $A[(i+2)\%N]$ etc. until the key-value pair is inserted.
- **Definition 1.2.5.8:** Quadratic probing iteratively tries the buckets $A[(i + f(j)) \% N]$ for $j = 0, 1, 2, \dots$ where $f(j) = j^2$ until finding an empty bucket. This avoids clustering patterns that occur with linear probing. However, it creates secondary clustering. This strategy may not find an empty bucket in A even if one exists.
- **Definition 1.2.5.9:** Using double hashing, we choose a secondary hash function h' , and if h maps some key k to a bucket $A[i]$, with $i = h(k)$, that is already occupied, then we iteratively try the buckets $A[(i + f(j)) \% N]$ for $j = 0, 1, 2, \dots$, where $f(j) = j \cdot h'(k)$. A common choice of secondary hash function is $h'(k) = q - (k \% q)$, for some prime number $q < N$. The secondary hash function should not evaluate to 0.

Definition 1.2.5.10: Deletions from the hash table must not hinder future searches. To ensure future searches are not hindered, we use tombstones. These are markers that are left in a bucket after a deletion. If a tombstone is encountered when searching along a probe sequence to find a key, we know to continue. The use of tombstones lengthens the average probe sequence distance. Two possible remedies are:

- **Definition 1.2.5.11:** Using local reorganisation, after deleting a key, we continue to follow the probe sequence of that key and move records into the vacated bucket.

- Or, we can periodically rehash the table by reinserting all records into a new hash table. If you have a record of which keys are accessed most, these can be placed where they can be found most easily.

2 Asymptotic Notation and Sorting

2.1 Asymptotic Notation

2.1.1 Types of Complexity

Definition 2.1.1.1: The time complexity of an algorithm can be expressed as the number of basic operations executed by an algorithm when computing for a set input size.

Definition 2.1.1.2: The space complexity of an algorithm is the amount of memory required to carry out the algorithm on an input of a set size.

Definition 2.1.1.3: The worst-case time complexity of an algorithm is the largest number of basic operations possible for an input of a set size.

2.1.2 General Notation

Definition 2.1.2.1: Generally, Big-O Notation denotes that, if $f = O(g)$, then f will grow no faster than g (the upper-bound for the growth rate, or $f' \leq g'$). Formally, let f be a real or complex function, and let g be a real function such that

$$f(x) = O(g(x)) \quad \text{as } x \rightarrow \infty \quad (1)$$

meaning $f(x)$ is Big-O $g(x)$ if the absolute value of $f(x)$ is at most a positive constant multiple of the absolute value of $g(x)$ for a suitably large value of x :

$$|f(x)| \leq C \cdot |g(x)| \quad x \geq x_0, \quad x_0 \in \mathbb{R}, \quad C \in \mathbb{N} \quad (2)$$

Definition 2.1.2.2: Similarly, Big-Omega Notation denotes that, if $f = \Omega(g)$, then f will grow at least as fast as g (the lower-bound for the growth rate, or $f' \geq g'$). Formally, let f and g be functions $\mathbb{R} \rightarrow \mathbb{R}$. We say that

$$f(x) = \Omega(g(x)) \quad (3)$$

if, for a suitably large value of x , there exists a constant C such that

$$|f(x)| \geq C \cdot |g(x)| \quad x \geq x_0, \quad x_0 \in \mathbb{R}, \quad C \in \mathbb{N} \quad (4)$$

Therefore, $f(x)$ is $\Omega(g(x))$ if $g(x)$ is $O(f(x))$.

Definition 2.1.2.3: Big-Theta Notation denotes the exact growth rate of a function. This means that, for some real-valued functions f and g ,

$$f(x) = O(g(x)) \quad \text{and} \quad f(x) = \Omega(g(x)) \quad (5)$$

Definition 2.1.2.4: Little-o Notation can be used for disregarding “small-order” terms. Generally, $f(x) = o(g(x))$ if $f'(x) < g'(x)$. It is formally defined for the real-valued functions f and g such that $f(x) = o(g(x))$ when

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0 \quad (6)$$

When the limit is removed, this becomes

$$o(g) = \{f : \mathbb{N} \rightarrow \mathbb{N} \mid \forall C > 0 \exists k > 0 : C \cdot f(n) < g(n) \forall n \geq k\} \quad (7)$$

Therefore, $f(x) = o(g(x)) \implies f(x) = O(g(x))$.

Definition 2.1.2.5: Similar to Little-o Notation, Little-omega Notation is defined such that $f = \omega(g)$ if $f' > g'$, and is defined formally as

$$\omega(g) = \{f : \mathbb{N} \rightarrow \mathbb{N} \mid \forall C > 0 \exists k > 0 : f(n) > C \cdot g(n) \forall n \geq k\} \quad (8)$$

Definition 2.1.2.6: A function is designated as a sublinear function if it grows slower than any linear function. Using Little-o Notation, $f(x)$ is sublinear if $f(x) = o(x)$, meaning

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = 0 \quad (9)$$

2.2 Sorting

2.2.1 Common Algorithms

Definition 2.2.1.1: An insertion sort works by taking the i^{th} element from a list and inserting it into an already sorted sequence of elements $[1, j - 1]$. This has a best-case runtime of $(n - 1)$ and a worst-case of $\frac{n(n-1)}{2}$, giving a Big-O of $O(n^2)$.

Definition 2.2.1.2: A selection sort takes the i^{th} smallest elements and places them in order at the start of the list until all elements have been sorted. Its time complexity can be calculated by doing the following:

$$\sum_{i=1}^{n-1} \sum_{j=i+1}^n 1 = \sum_{i=1}^{n-1} (n - i) = \left(\sum_{i=1}^{n-1} n \right) - \left(\sum_{i=1}^{n-1} i \right) = n(n-1) - \frac{n(n-1)}{2} = O(n^2) \quad (10)$$

Definition 2.2.1.3: A bubble sort takes the i^{th} largest elements and places them in order at the end of the list until all elements have been sorted. Its time complexity can be calculated by doing the following:

$$\sum_{i=1}^{n-1} (n - i) = O(n^2) \quad (11)$$

Definition 2.2.1.4: A merge sort takes a list of elements and recursively splits it into two even parts until each sublist contains one element. Once this occurs, the sublists are merged back together in the correct order to form the sorted list. It does this with a variant of an insertion sort. It has a time complexity of $O(n \log n)$.

Definition 2.2.1.5: A quick sort takes the first, middle, and final element of a list and selects one as a pivot point. The remaining elements are then placed in sublists depending on whether they are smaller/larger than the pivot. This gives a best-case time complexity of $O(n \log n)$ and a worst-case of $O(n^2)$.

2.2.2 Recursion for Sorting and Solving Recurrences

To analyse recursive algorithms, first of all note that most of them are actually hybrids between iterative and strictly recursive. For example, merge sort splits the input of size n into two halves of equal size $\frac{n}{2}$, independently recurses in each half, then merges the resulting sorted sequence.

$$T(n) \leq \begin{cases} d & \text{if } n \leq c \text{ for constants } c, d > 0 \\ 2 \cdot T\left(\frac{n}{2}\right) + a \cdot n & \text{otherwise} \end{cases} \quad (12)$$

Definition 2.2.2.1: The Master Theorem can be used to quickly solve recurrences of form

$$T(n) \leq aT\left(\frac{n}{b}\right) + f(n) \quad a \geq 1, b > 1 \quad (13)$$

The theorem has three cases:

1. $f(n) = O(n^{\log_b a - \epsilon})$, $\epsilon > 0 \implies T(n) = \Theta(n^{\log_b a})$
2. $f(n) = \Theta(n^{\log_b a} \cdot (\log n)^k) \implies T(n) = \Theta(n^{\log_b a} \cdot (\log n)^{k+1})$
3. $f(n) = O(n^{\log_b a + \epsilon})$, $\epsilon > 0$ and $af\left(\frac{n}{b}\right) \leq cf(n)$, $c < 1 \implies T(n) = \Theta(f(n))$

3 Selection and Data Structures

3.1 Algorithms for Sorting, Searching, and Selection

3.1.1 Sorting

Definition 3.1.1.1: A bucket sort algorithm creates buckets with labels in the interval $[1, k]$ where k is the largest value in the list. Each element is then placed in its corresponding bucket before the buckets are outputted in ascending order, creating a sorted list. This gives a worst-case time complexity of $O(n + k)$. If k is small relative to n , this gives a best-case time complexity of $o(n \log n)$.

Definition 3.1.1.2: A bucket sort recursively sorts a list of items by a given digit of each element until the list is sorted (from right-most digit to left-most). This algorithm requires as many buckets as the base of the number system being used (e.g. binary requires two buckets). It has a time complexity of $O(n \log_d k)$ for a list of values in $[1, k - 1]$ in base d .

3.1.2 Searching

Definition 3.1.2.1: A binary search assumes that the list being searched is sorted. It recursively halves the search space on each pass until the desired element is found. This produces the recurrence

$$T(n) = T\left(\frac{n}{2}\right) + O(1) = O(\log n) \quad (14)$$

3.1.3 Selection

Selection algorithms are used to, for example, find the i^{th} smallest element in an unsorted list.

Definition 3.1.3.1: The quick select algorithm works similarly to the quick sort, using pivots to split the search space and reduce execution times. In the worst-case, the algorithm is $O(n^2)$, but on average executes in $O(n)$ time.

Definition 3.1.3.2: The median-of-medians algorithm is doubly recursive and works by calculating the median of the list and using it as a pivot. This gives the recurrence

$$T(n) = T\left(\frac{n}{5}\right) + T\left(\frac{7n}{10} + 3\right) + n = O(n) \quad (15)$$

3.2 Trees

3.2.1 Binary Search Trees

Definition 3.2.1.1: A rooted binary tree is a tree where each node has a single parent node and at most two children nodes. The binary tree is rooted if there is one node with all other nodes as descendants.

Tree Properties:

- **Definition 3.2.1.2:** A tree's size is the total number of nodes in the tree.
- **Definition 3.2.1.3:** A node's depth is its distance from the root node.

Definition 3.2.1.4: A binary tree becomes a binary search tree if, for all nodes, all left-descendant nodes are smaller than it, and all right-descendant nodes are greater than it.

Theorem 3.2.1.1: Let T be a rooted binary tree of height h . Then,

- T has at most $(2^{h+1} - 1)$ nodes.
- T has at most 2^h leaves.

Proof: The maximum number of nodes is in a complete tree of height h , meaning all levels are full,

$$1 + 2 + 2^2 + 2^3 + \dots + 2^h = \sum_{n=0}^h 2^n = 2^{h+1} - 1 \quad (16)$$

The second statement follows by induction on h . Case $h = 0$ is obvious. We then consider the left and right subtrees of the root of T . Then each of them has height $\leq (h - 1)$, and (by induction hypothesis) $\leq 2^{h-1}$ leaves. Then T has at most $2 \times 2^{h-1} = 2^h$ leaves.

□

Trees can be traversed in one of three ways, these are:

- **Definition 3.2.1.5:** Preorder traversal where the root is visited, followed by the left and right subtrees.
- **Definition 3.2.1.6:** Inorder traversal where the root is visited in between the left and right subtrees.
- **Definition 3.2.1.7:** Postorder traversal where the root is visited after both the left and right subtrees.

3.2.2 Balanced Binary Search Trees

Definition 3.2.2.1: Balanced binary search trees are binary search trees which balance themselves using rotation operations to minimise height differences between subtrees.

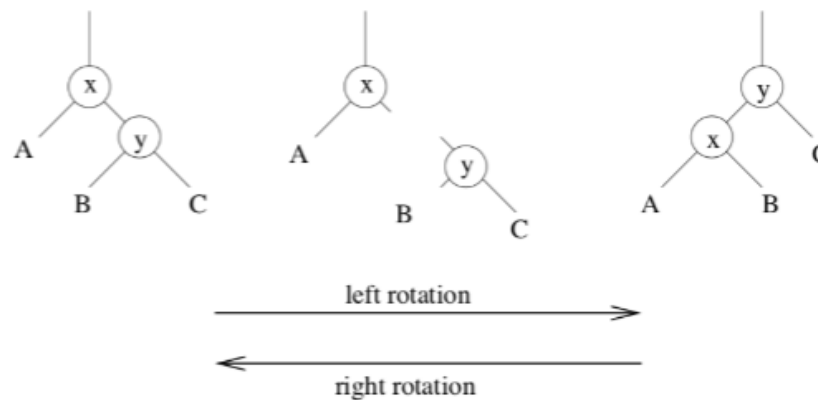


Figure 1: The basic rotation operations.

3.2.3 AVL Trees

Definition 3.2.3.1: An AVL tree is a self-balancing binary search tree with the height-balance property (for each node, the height of its children differ by at most one).

Theorem 3.2.3.1: The height of an AVL tree storing n nodes is $O(\log n)$

Proof: Let us bound $n(h)$ as the minimum number of internal nodes of an AVL tree of height h . We can clearly see that $n(1) = 1$ and $n(2) = 2$. For $n > 2$, an AVL tree of height h contains the root node, one AVL subtree of height $n(h - 1)$, and another of height $n(h - 2)$. That is,

$$n(h) = n(h - 1) + n(h - 2) \quad (17)$$

Knowing that $n(h - 1) > n(h - 2)$, we get $n(h) > 2n(h - 2)$ so, by induction,

$$n(h) > 2^i n(h - 2i) \quad (18)$$

Taking i to be $\frac{h}{2} - 1$, so that $h - 2i = 1$ or 2 , on solving we get,

$$n(h) > 2^{\frac{h}{2}-1} \quad (19)$$

Taking logarithms, $h < 2\log(n(h)) + 2$. Since $\log(n) > \log(n(h))$, the height of an AVL tree is $O(\log n)$. □

Definition 3.2.3.2: The algorithm for AVL insertion automatically rebalances the tree after inserting a new element to uphold the height-balance property. For example, for a new node n ,

- If it was inserted into the left subtree of a left child, the algorithm performs a single right rotation.
- If it was inserted into the right subtree of a right child, the algorithm performs a single left rotation.
- If it was inserted into the right subtree of a left child, the algorithm performs a left rotation followed by a right rotation.
- If it was inserted into the left subtree of a right child, the algorithm performs a right rotation followed by a left rotation.

3.3 Heaps

3.3.1 Heap Properties

Definition 3.3.1.1: A priority queue is a collection of prioritised elements that allows arbitrary element insertion and removal having the first priority (highest or lowest).

Definition 3.3.1.2: All heaps are complete binary trees. The first item in a heap is the root, the second is the left child, and the third is the right child. The next nodes always fill the next levels left-to-right.

Definition 3.3.1.3: A max heap stores values where, for every node except the root, the value is at most that of its parent. In any subtree, no values are larger than the value stored in the subtree root.

Definition 3.3.1.4: A min heap stores values where, for every node except the root, the value is at least that of its parent. In any subtree, no values are smaller than the value stored in the subtree root.

Definition 3.3.1.5: Assuming that we start counting at one for the root, the heap indices can be calculated such that, for a node at index i , the left child is at $2i$, the right child is at $2i + 1$, and the parent is at $\lfloor \frac{i}{2} \rfloor$.

3.3.2 Heap Operations

Definition 3.3.2.1: Heapify takes an incorrectly ordered heap and reorders it into an acceptable order. It has a runtime of $O(\log n)$.

Algorithm 2 Heapify

Input: A , the array to turn into a heap, v , the index of the largest value, i , the highest index.

Output: A correctly ordered heap.

```

largest  $\leftarrow v$ 
leftChild  $\leftarrow 2v$ 
rightChild  $\leftarrow 2v + 1$ 

if leftChild  $\leq i$  and  $A[\text{leftChild}] > A[\text{largest}]$  then
    largest  $\leftarrow \text{leftChild}$ 
end if
if rightChild  $\leq i$  and  $A[\text{rightChild}] > A[\text{largest}]$  then
    largest  $\leftarrow \text{rightChild}$ 
end if

if largest  $\neq v$  then
    Swap( $A[v]$ ,  $A[\text{largest}]$ )
    Heapify( $A$ , largest,  $i$ )
end if

```

Definition 3.3.2.2: Insertion adds the new element to the bottom-left of the heap before running Heapify.

Definition 3.3.2.3: Build heap takes an array of numbers and converts it into a heap by running Heapify iteratively, starting v as the size of the array and looping down to one.

Theorem 3.3.2.4: The build heap algorithm runs in $O(n)$ time.

Proof: Let $T(n)$ be the execution time for a heap of size n .

$$T(n) = \sum_{h=1}^{\log n} (\# \text{ of nodes at height } h) \cdot O(h) \quad (20)$$

Given the properties of binary trees,

$$T(n) \leq \sum_{h=1}^{\log n} \frac{n}{2^h} \cdot O(h) = O\left(n \cdot \sum_{h=1}^{\log n} \frac{h}{2^h}\right) \quad (21)$$

This is well known for $x \in \{0, 1\}$,

$$\sum_{i=0}^{\infty} i \cdot x^i = \frac{x}{(1-x)^2} \quad \sum_{h=1}^{\log n} \frac{h}{2^h} \leq \sum_{h=0}^{\infty} h \cdot \left(\frac{1}{2}\right)^h = 2 \quad (22)$$

and hence,

$$T(n) = O\left(n \cdot \sum_{h=1}^{\log n} \frac{h}{2^h}\right) = O(n) \quad (23)$$

□

3.4 Lower Bounds for Sorting and Selection

Theorem 3.4.0.1: Any comparison-based sorting algorithm requires $\Omega(n \log n)$ comparisons in the worst case.

Proof: It is sufficient to determine the minimum height of a decision tree in which each permutation appears as a leaf. We can consider a decision tree of height h with l leaves corresponding to a comparison sort on n elements. Each of the $n!$ permutations of the input appears as some leaf, $l \geq n!$. Given that a binary tree of height h has at most 2^h leaves, we get that

$$n! \leq l \leq 2^h \quad (24)$$

Therefore, $n! \leq 2^h$, and after taking logs of both sides,

$$h \geq \log(n!) \equiv \Omega(n \log n) \quad (25)$$

□

Definition 3.4.0.1: An adversary algorithm is designed to intercept and limit access to the input by the original algorithm such that the time taken to make a decision is maximised. However, the adversary does not have knowledge of how the algorithm works, so must be general enough to work for any algorithm provided.

Theorem 3.4.0.2: Any comparison-based algorithm attempting to find the maximum element in an unsorted array requires at least $(n - 1)$ comparisons in the worst case.

Proof: After at most $(n - 2)$ comparisons no more than two elements can have not lost a comparison. An adversary can make either of them max and be consistent, but there is not enough information for the original algorithm to be consistent. Hence, the algorithm needs to make one more comparison, making the minimum number of comparisons $(n - 1)$.

□

Alternative Proof: Taking this into account, we can design a strategy for the adversary. The only relevant information for the algorithm is how many elements have lost at least one comparison. As soon as $(n - 1)$ elements have lost, the algorithm can decide that the only undefeated element must be the maximum.

Assuming each index i has a status (N for never lost, and L for lost), initially all indices have N . After each comparison there can be a status change $N \rightarrow L$. Therefore, the adversary must delay changes $N \rightarrow L$ as much as possible.

Given this information, each comparison will either update one status from N to L , or will update neither, providing the algorithm with at most one bit of new information. In total, the algorithm needs $(n - 1)$ bits of information to make a decision, and hence at least $(n - 1)$ comparisons are necessary.

□

4 Graph Theory and Algorithms

4.1 The Basics of Graph Theory

4.1.1 Types of Graphs

Definition 4.1.1.1: A graph G is a pair $(V(G), E(G))$ of sets where $V(G)$ is a non-empty set of vertices making up the graph and $E(G)$ is a set of unordered pairs, $\{u, v\}$, $u, v \in G$.

Definition 4.1.1.2: A directed graph is a graph where the edge set $E(G)$ instead contains ordered pairs, indicating the direction of travel along each edge.

Definition 4.1.1.3: A multi-graph allows for multiple edges to exist between vertices, meaning the edge set may contain duplicates.

Definition 4.1.1.4: A pseudo-graph allows loops, meaning pairs in the edge set can contain two instances of the same vertex.

Definition 4.1.1.5: A vertex/edge-weighted graph assigns a weight for each vertex or edge.

Definition 4.1.1.6: A complete bipartite graph is a graph such that any subgraph is a bipartite graph, meaning we can separate its vertex set into two separate sets where each edge has an endpoint in both sets.

Definition 4.1.1.7: A complete graph contains all possible edges between vertices.

Definition 4.1.1.8: The hypercube, n-dimensional cube, or n-cube is the graph with

$$V = \{(e_1, \dots, e_n) | e_i \in \{0, 1\} (i = 1, \dots, n)\} \quad (26)$$

in which two vertices are neighbours iff the corresponding rows differ in exactly one entry.

4.1.2 Other Terminology

Let G be a graph and uv be an edge in it. Then,

- **Definition 4.1.2.1:** u and v are called endpoints of the edge uv .
- **Definition 4.1.2.2:** u and v are called neighbours or adjacent vertices.
- **Definition 4.1.2.3:** uv is said to be incident to u and v .
- **Definition 4.1.2.4:** If uv is also an edge where $v \neq w$, then uv and uw are adjacent edges.

Definition 4.1.2.5: The neighbourhood of a vertex $v \in V$, denoted by $N(v)$, is the set of neighbours of v .

Definition 4.1.2.6: The degree of a vertex $v \in V$, $\deg(v)$, is the size of its neighbourhood, $|N(v)|$.

Definition 4.1.2.7: $\delta(G)$ and $\Delta(G)$ give the vertices with the smallest and largest degrees, respectively.

Definition 4.1.2.8: A vertex with degree zero is an isolated vertex.

Definition 4.1.2.9: A vertex with degree one is an end vertex or a pendant vertex.

Definition 4.1.2.10: A subgraph $G' = (V', E')$ of $G = (V, E)$ is a graph with $V' \subseteq V$ and $E' \subseteq E$.

- **Definition 4.1.2.11:** A subgraph is proper if $G' \neq G$.
- **Definition 4.1.2.12:** A subgraph is spanning if $V' = V$.
- **Definition 4.1.2.13:** An induced subgraph is a subgraph where E' contains all edges of E between vertices of V' . It is obtained by removing from G all vertices of $G \setminus G'$.

4.1.3 Basic Theorems

Theorem 4.1.3.1: Known as the Handshaking Lemma, if $G = (V, E)$ is a graph, then $\sum_{v \in V} \deg(v) = 2|E|$.

Proof: Every edge has two endpoints and contributes one to each of their degrees, so contributes two to the sum of the degrees of all vertices of V .

□

Theorem 4.1.3.2: In every undirected graph, the number of vertices of odd degree is even.

Proof: Let $G = (V, E)$ and $V_{\text{odd}} = \{v : \deg(v) \text{ is odd}\}$ and $V_{\text{even}} = \{v : \deg(v) \text{ is even}\}$. Clearly $\sum_{v \in V_{\text{even}}} \deg(v)$ is even. By the Handshaking Lemma,

$$\sum_{v \in V_{\text{odd}}} \deg(v) = 2 \cdot |E| - \sum_{v \in V_{\text{even}}} \deg(v) \quad (27)$$

it follows that $\sum_{v \in V_{\text{odd}}} \deg(v)$ is even too. Now, if we have an odd number of vertices of odd degree, then there is a contradiction, meaning there must be an even number of vertices with an odd degree.

□

Theorem 4.1.3.3: All n -dimensional cubes are bipartite.

Proof: Let K_n be an n -dimensional hypercube. Create two vertex sets, V_1 and V_2 , containing all the vertices of K_n with even and odd numbers of ones, respectively. Due to the definition of a hypercube, it is guaranteed for each edge to have an endpoint in both of the resulting vertex sets. Therefore, the n -cube must be bipartite.

□

4.2 Paths, Cycles, and Connectivity

4.2.1 Walks, Paths, and Cycles

Definition 4.2.1.1: A walk is a sequence of edges from a graph G .

Definition 4.2.1.2: A path in a graph is a subgraph which is isomorphic to the graph P_k for some integer $k \geq 1$.

Definition 4.2.1.3: A circuit is a walk where the starting node is also the ending node.

Definition 4.2.1.4: A cycle is a subgraph isomorphic to the graph C_k where $k \geq 3$.

Definition 4.2.1.5: The length of a path is the number of edges within it.

Definition 4.2.1.6: The distance between two nodes, $\text{dist}(u, v)$, is the length of the shortest path between the two vertices, and is defined as ∞ if no path exists.

Definition 4.2.1.7: The diameter of a graph is the largest possible distance between two vertices in the graph.

4.2.2 Connectivity

Definition 4.2.2.1: A graph is connected if, between every pair of vertices, there exists at least one path.

Definition 4.2.2.2: A connected component of a graph is a maximally connected subgraph.

Theorem 4.2.2.1: Every graph contains at least $|V| - |E|$ connected components.

Proof: We can prove this by induction over $m = |E|$. When $m = 0$ there are no edges, only isolated vertices, meaning there must be $|V|$ connected components.

Let $G = (V, E)$ with $|E| = m + 1$. Consider an arbitrary $e \in E$, and define $E' = E \setminus \{e\}$. By the inductive hypothesis, $G' = (V, E')$ has at least $|V| - |E'| = |V| - m$ connected components. This produces two cases: either, with e , the number of connected components does not change; or, with e , the number of connected components decreases by one. In both cases, G has at least $|V| - m - 1 = |V| - |E|$ connected components, proving the initial hypothesis.

□

Definition 4.2.2.3: A directed graph is weakly connected if the graph obtained from G by forgetting directions is connected.

Definition 4.2.2.4: A directed graph is strongly connected if any two distinct vertices are connected by paths in both directions.

Definition 4.2.2.5: A strongly connected component of a directional graph is a maximal strongly connected subgraph.

4.2.3 Special Circuits and Cycles in Graphs

Definition 4.2.3.1: A Eulerian circuit is a graph where we can start and end at the same node while traversing each edge exactly once.

Definition 4.2.3.2: A Hamiltonian cycle is a graph where we can start and end on the same node while visiting each node exactly once.

Theorem 4.2.3.1: A connected graph with at least two vertices has an Eulerian circuit iff each of its vertices has even degree.

Proof: This is proved in (Rosen, 2003, pp.634-636).

□

4.3 Trees and Isomorphism

4.3.1 Trees

Definition 4.3.1.1: A forest is an acyclic graph.

Definition 4.3.1.2: A tree is a connected forest.

Theorem 4.3.1.1: Every connected graph contains a spanning tree.

Proof: Let G be a connected graph. If G doesn't contain cycles then it is a tree and therefore is its own spanning tree. Conversely, if the graph does contain a cycle, then we can remove one edge from the graph, breaking the cycle and forming a spanning tree. The resulting graph will still be connected as cycles will contain at least one vertex of degree two, meaning removing an edge won't produce an isolated vertex.

□

Definition 4.3.1.3: A leaf is a vertex of degree one.

Theorem 4.3.1.2: Every tree on at least two vertices contains a leaf.

Proof: By contradiction, we assume that every vertex has degree zero or two. If a vertex has degree zero, then the graph is not connected, and therefore cannot be a tree. Similarly, if every vertex has degree two then we can traverse through the tree by travelling to each vertex's neighbours until we reach a node that has already been visited, proving a cycle exists, making the graph not a tree. Therefore, trees with at least two vertices must contain at least one leaf.

□

Alternative Proof: (When every vertex has a degree of at least two). Consider a longest path P and an end vertex v of P . All neighbours of v are on P as the tree must be connected, therefore if $\deg(v) \geq 2$, then there is a cycle, meaning a tree cannot exist. This holds for both endpoints of the path, proving there must be at least one vertex of degree one for the graph to be a tree.

□

Theorem 4.3.1.3: A connected graph on n vertices is a tree iff it has $(n - 1)$ edges.

Proof: For small values of n , the theorem holds, e.g., a tree with one vertex has no edges, and a tree with two vertices has one edge. Suppose each tree on $(n - 1)$ vertices has $(n - 2)$ edges. Take a tree on $n \geq 3$ vertices, T contains a leaf v . Consider the graph $T - v$, it has one vertex less and one edge less than T . $T - v$ is still acyclic and such is a tree with $n - 1$ vertices, by the inductive hypothesis it has $(n - 2)$ edges. T has one edge more, so $(n - 1)$ edges.

□

Theorem 4.3.1.4: Let T be a tree and $u, v \in V(T)$ with $u \neq v$. Then there is a unique path in T between u and v .

Proof: By contradiction, suppose there are two paths P and Q in T between u and v . Let x and y in $V(T)$ be distinct and chosen in such a way that x and y are on P and Q , but between x and y the vertices of P and Q are disjoint. Then the segments of P and Q form a cycle, contradicting that T is a tree.

□

Definition 4.3.1.4: A rooted tree is a tree in which one vertex is fixed as the root (and every other vertex is directed away from the root).

Let v be a vertex in a rooted tree T .

- **Definition 4.3.1.5:** The neighbours of v in the next level are called the children of v .
- **Definition 4.3.1.6:** The (unique) neighbour of v in the previous level is called the parent of v .
- **Definition 4.3.1.7:** If v has children, then it is an internal vertex.

Theorem 4.3.1.5: A tree with at least two vertices, n_3 of which have degree at least three, has at least $(n_3 + 2)$ leaves.

Proof: Let T be a tree on $n \geq 2$ vertices. We use induction on n . Let $l(t)$ denote the number of leaves in T , and $n_3(T)$ denote the number of vertices of degree at least three in T . If $n = 2$, $n_3 = 0$ and T has two leaves. Now suppose that every tree on fewer than n vertices has at least $(n_3 + 2)$ leaves, and consider a tree T on $n \geq 3$ vertices. Since T is a tree on at least three vertices, it has a leaf u . Then $T' = T - u$ is a tree on $(n - 1)$ vertices. By the induction hypothesis, we have

$$l(T') \geq n_3(T') + 2 \tag{28}$$

Let v be the unique neighbour of u in T . T is connected and has at least three vertices, so v has at least two neighbours in T . This results in one of three cases:

1. Suppose that v has exactly two neighbours in T . Then $n_3(T) = n_3(T')$ and $l(T) = l(T')$. Hence,

$$l(T) = l(T') \geq n_3(T') + 2 = n_3(T) + 2 \tag{29}$$

2. Suppose that v has exactly three neighbours in T . Then $n_3(T) = (n_3(T') + 1)$ and $l(T) = (l(T') + 1)$. Hence,

$$l(T) = l(T') + 1 \geq n_3(T') + 2 + 1 = n_3(T) - 1 + 2 + 1 = n_3(T) + 2 \tag{30}$$

3. Suppose that v has at least four neighbours in T . Then $n_3(T) = n_3(T')$ and $l(T) = (l(T') + 1)$. Hence,

$$l(T) = l(T') + 1 \geq n_3(T') + 2 + 1 = n_3(T) + 2 + 1 \geq n_3(T) + 2 \tag{31}$$

In each case, $l(T)$ is at least $n_3 + 2$.

□

Theorem 4.3.1.6: Every tree is a bipartite graph.

Proof: Let T be a tree and u be any one of its vertices. Place v in the set V_1 . For every vertex $u \neq v$, there is a unique path from v to u in T , consider the length of this path. If the length is odd, put u in V_2 , otherwise, place it in V_1 . V_1 and V_2 are disjoint and together make up $V(T)$ and every edge of T will have an endpoint in both V_1 and V_2 .

□

4.3.2 Isomorphism

Definition 4.3.2.1: Two graphs $G(V, E)$ and $G'(V', E')$ are isomorphic if there exists a bijective function $f : V \rightarrow V'$ such that for every $u, v \in V$, we have $uv \in E$ iff $f(u)f(v) \in E'$. Then we write $G \cong G'$.

Definition 4.3.2.2: Two rooted trees $T_1(V_1, E_1, r_1)$ and $T_2(V_2, E_2, r_2)$ are called isomorphic rooted trees if there exists an isomorphism bijection $f : V_1 \rightarrow V_2$ such that $f(r_1) = r_2$.

Definition 4.3.2.3: In a rooted tree with root r , the level, $L(i)$ is the set of vertices at distance i from the root r .

Algorithm 3 Label Vertex

Input: T , a tree, and v , a vertex from the tree.

Output: A label for vertex v .

```

if  $v$  is a leaf of  $T$  then
     $label(v) \leftarrow 10$ 
else
    for every child  $w$  of  $v$  do
         $l(w) \leftarrow LabelVertex(T, w)$ 
    end for
    Sort the labels of the children decreasingly.
     $label(v) \leftarrow 1$ 
end if
return  $label(v)$ 

```

Algorithm 4 Labelled Tree Isomorphism

Input: Two pairs of trees and roots, $(T_i, r_i) \ i \in \{1, 2\}$.

Output: YES or NO.

```

 $label(r_1) \leftarrow LabelVertex(T_1, r_1)$ 
 $label(r_2) \leftarrow LabelVertex(T_2, r_2)$ 
if  $label(r_1) = label(r_2)$  then
    return YES
else
    return NO
end if

```

Theorem 4.3.2.1: If (T_1, r_1) is isomorphic to (T_2, r_2) then Algorithm 4 returns YES.

Proof: We use induction on the number k of levels. When $k = 1$, the two trees have only their root (thus must be isomorphic), which gets label 10. Thus the algorithm returns YES. If two rooted trees are isomorphic and both have k levels then the algorithm returns YES on input. Let (T_1, r_1) and (T_2, r_2) be two isomorphic rooted trees, each with $(k + 1)$ levels.

Let f be the bijection between T_1 and T_2 . Then $r_2 = f(r_1)$ and, for every child a_1 of r_1 there exists a child a_2 of r_2 such that the subtrees rooted at a_i are isomorphic, and $a_2 = f(a_1)$. Since every such pair of subtrees are isomorphic, they both have k levels. Thus, by the induction hypothesis, Algorithm 3 returns the same label for their roots. Since all these labels are the same, Algorithm 4 returns YES.

□

4.4 Breadth and Depth First Searches

4.5 Minimum Spanning Trees

4.5.1 Introduction to Minimum Spanning Trees

Definition 4.5.1.1: The minimum spanning tree problem tries to find a tree that spans all the vertices in a graph for the lowest possible cost. Minimum spanning trees have $(|V| - 1)$ edges, have no cycles, and might not be unique.

Definition 4.5.1.2: Kruskal's algorithm attempts to solve the problem by:

1. Sort the edges by weight.
2. Let $A = \emptyset$
3. Consider the edges in increasing order of weight. For each edge e , add e to A unless this would create a cycle.

It has a running time of $O(E \log V)$.

Definition 4.5.1.3: Prim's algorithm attempts to solve the problem by:

1. Let $U = \{u\}$ where u is some vertex chosen arbitrarily.
2. Let $A = \emptyset$
3. Until U contains all vertices: find the least-weight edge e that joins a vertex v in U to a vertex w not in U and add e to A and w to U .

It has a running time of $O(V \log V + E)$.

Definition 4.5.1.4: Let A be a subset of an MST. If the set $A \cup \{(u, v)\}$ is also a subset of an MST, then (u, v) is a safe edge for A .

Definition 4.5.1.5: A cut $(S, V - S)$ of $G = (V, E)$ is a partition of V .

Definition 4.5.1.6: An edge $(u, v) \in E$ crosses a cut $(S, V - S)$ if $u \in S$ and $v \in V - S$, or vice-versa.

Definition 4.5.1.7: An edge $(u, v) \in E$ is a light edge crossing a cut if its weight is smallest among all the edges crossing the cut.

Definition 4.5.1.8: A cut respects a set A of edges if no edge of A crosses the cut.

Algorithm 5 Generic MST Algorithm

Input: A graph G .

Output: An MST.

$A \leftarrow \emptyset$

while A does not form a spanning tree **do**

Find an edge (u, v) that is safe for A

$A \leftarrow A \cup \{(u, v)\}$

end while

return A

4.5.2 Implementing MST Algorithms

Algorithm 6 Implementation of Kruskal's Algorithm

Input: The set of vertices V and the set of edges E .

Output: An MST.

```

 $A \leftarrow \emptyset$ 
for Each vertex  $v \in V$  do
     $C(v) \leftarrow v$ 
end for
Sort  $E$ 
while  $E \neq \emptyset$  do
     $e \leftarrow (u, v) \in E$  with min cost
    if  $C(u) \neq C(v)$  then
         $A \leftarrow (A + e)$ 
        for Each vertex  $w \in C(u), w \in C(v)$  do
            Update  $C(w)$  with  $C(u) \cup C(v)$ 
        end for
    end if
end while
return  $A$ 
  
```

The sorting initially takes $O(E \log V)$ time, the union-find operations also take $O(E \log V)$ time, so the total running time is $O(E \log V)$.

Algorithm 7 Implementation of Prim's Algorithm

Input: The set of vertices V and the set of edges E .

Output: An MST.

```

 $U = \{u\}$ 
 $A \leftarrow \emptyset$ 
for Each vertex  $v \in (V - u)$  do
     $B(v)$  is the least-weight edge from  $v$  to  $U$ 
end for
while  $U \neq V$  do
    Choose  $v$  with minimum cost  $B(v)$ 
     $A \leftarrow (A + e)$ 
     $U \leftarrow (U + v)$ 
    Update  $B$ 
end while
return  $A$ 
  
```

To initialise the algorithm, all edges are considered. It then iterates through the while loop once per vertex extracting the minimum cost edge and performing updates in $O(\log V)$ time. This gives an overall running time of $O(V \log V + E)$.

Part II

Computational Thinking

5 Introduction to Computer Science

6 Logic

Part III

Computer Systems

7 Digital Electronics

7.1 Circuits

7.1.1 Transistors

Definition 7.1.1.1: Transistors are electronically controlled switches that turn on or off when voltage or current is applied to a control terminal.

Definition 7.1.1.2: A nMOS transistor is a sandwich of layers of conducting and insulating material. The gate is a conducting layer, on top of an insulating layer, on top of the substrate which is a semi-conductor. When its gate g is at a high voltage, current is allowed to flow from source to drain, and this is prevented when the gate is at a low voltage.

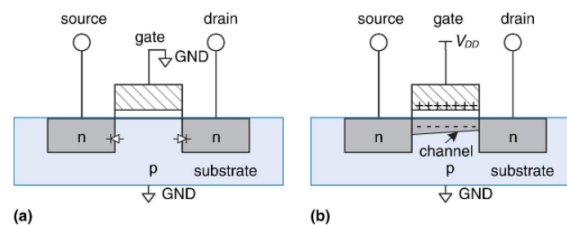


Figure 2: An nMOS transistor

The junction between p-type and n-type silicon is a diode. This means the current can only flow from p-type to n-type as they act as anodes and cathodes respectively.

Definition 7.1.1.3: A pMOS transistor is the opposite of a nMOS transistor - it allows current to flow when the gate is at low voltages, and prevents current flow at high voltages.

Definition 7.1.1.4: The static discipline states that if inputs to transistors meet valid thresholds, then the system must guarantee outputs will meet valid output thresholds and will not fall into the forbidden zone (outside of the accepted range for high/low voltages).

Definition 7.1.1.5: Moore's Law is usually quoted as “transistor density doubles every two years”, or “computer processing power doubles every eighteen months”.

7.1.2 Circuit Design

Definition 7.1.2.3: Abstraction is used when designing circuits to simplify problems by hiding unimportant details, highlighting the actual logic problem to be solved.

Definition 7.1.2.4: Another design principle used is discipline. Here, design choices are deliberately restricted to make circuits easier to model, design, and combine.

Definition 7.1.2.5: When designing complex systems, engineers use the three-y's: hierarchy, dividing a system into clear modules and submodules; modularity, using well-defined functions and interfaces for modules; and regularity, encouraging uniformity so modules can be swapped or reused.

7.1.3 Combinatorial Circuits

Definition 7.1.3.1: A combinatorial circuit is a circuit where the output depends purely on the current input.

Definition 7.1.3.2: An adder circuit takes the contents of two registers and adds them together.

- **Definition 7.1.3.3:** A half-adder circuit takes two one-bit binary inputs and adds them, returning a sum bit, and a carry bit.
- **Definition 7.1.3.4:** A full-adder circuit takes three one-bit binary inputs (one of which is a carry bit) and adds them, returning a sum bit, and a carry bit.

Definition 7.1.3.5: A decoder circuit takes an n -bit binary number and decodes it into 2^n data lines.

Definition 7.1.3.6: A multiplexor (mux) circuit uses a binary number to select one of a set of inputs to funnel to its output.

Definition 7.1.3.7: A tristate takes in two bits, an input bit and an enable bit. If the enable bit is a one, the input bit is outputted, if not then the output is left floating.

7.1.4 Sequential Circuits

Definition 7.1.4.1: A sequential circuit is a circuit which changes its output depending on its input history, alongside the current input.

Definition 7.1.4.2: A circuit which has two stable output states for a given input is known as a bistable circuit.

Definition 7.1.4.3: An sr-latch takes in two inputs, set S and reset R . If set has a pulse of one, then the output remains at one, even after the pulse ends. If reset has a pulse of one, then the output is reset back to zero. This is a bistable circuit.

Definition 7.1.4.4: A d-latch has two inputs, a data input D and a clock input CLK . The latch only changes to match the input at D when the CLK signal is high.

Definition 7.1.4.5: A d-type flip-flop works similarly to the d-latch, but only allows the stored output to change on the rising edge of the clock signal.

7.2 Introduction to Logic

7.2.1 Functionally Complete Sets

Definition 7.2.0.1: A functionally complete set is a set of boolean operators which can be used to express all possible truth tables by combining members of the set into a boolean expression. For example, $\{AND, OR, NOT\}$, $\{NOR\}$, and $\{NAND\}$ are all functionally complete sets.

8 Machine Architecture

8.1 MIPS

8.1.1 Introduction to MIPS

Definition 8.1.1.1: A MIPS (Microprocessor without Interlocked Pipeline Stages) is 32-bit RISC processor used in embedded systems and consumer electronics. It uses 32-bit words, has around 110 instructions, and 32 general purpose registers.

Definition 8.1.1.2: Amdahl's Law states that the performance improvement to be gained from using some faster mode of execution is limited by a fraction of the time the faster mode can be used.

8.1.2 The MIPS Instruction Set

The instruction set uses five addressing modes, these are:

- **Definition 8.1.2.1:** When all addresses in the instruction are registers, this is register-only addressing.
- **Definition 8.1.2.2:** When a value is used directly in an instruction, this is immediate addressing.
- **Definition 8.1.2.3:** Base addressing is when the address of a memory operand is given by a base address and a signed immediate.
- **Definition 8.1.2.4:** PC-relative addressing takes the current position in memory and jumps a given amount to a new address.
- **Definition 8.1.2.5:** Pseudo-direct addressing is used by MIPS in place of direct addressing as it does not support 32-bit addresses. As a result, it calculates it from 26 as the two least significant bits are always zeros as all words are multiples of four, and the four most significant bits can also be left as zero.

The MIPS instruction set can be split into three instruction types, these are:

- **Definition 8.1.2.6:** R-type instructions contain a six-bit opcode, three five-bit registers (two source and one destination), a five-bit shift value, and a six-bit code to identify the function.
- **Definition 8.1.2.7:** I-type instructions contain a six-bit opcode, two five-bit registers (one source, one destination), and a sixteen-bit immediate.
- **Definition 8.1.2.8:** J-type instructions contain a six-bit opcode and a twenty-six bit memory address. The compiler combines the two to form a new thirty-two bit address.

8.1.3 Memory Maps

MIPS uses 32-bit memory addresses across 2^{32} bits (or 4 GB), ranging from 0x0 to 0xFFFFFFFFC. This address space is split into the four segments:

- **Definition 8.1.3.1:** The text segment stores the machine language program and can store 256 MB.
- **Definition 8.1.3.2:** The global data segment stores global variables for program functions to access. These can be accessed using the pointer `$gp`. This segment can store 64 KB of data.
- **Definition 8.1.3.3:** The dynamic data segment stores data which is dynamically allocated through the execution of the program and can store up to 2 GB of data using a stack and a heap.
- **Definition 8.1.3.4:** The reserved segment is used by the operating system and cannot be directly accessed by the program.

8.2 The ATmega328P Microcontroller

8.2.1 Main Technical Characteristics

Definition 8.2.1.1: A microcontroller is an integrated device consisting of a microprocessor, memory, and an input/output. They are used in embedded systems and store their program in flash memory.

The microcontroller has 32 general purpose 8-bit registers that are copied onto SRAM memory. Registers 26-31 are used in addressing, and as such should be treated as three 16-bit registers, rather than six 8-bit registers.

Definition 8.2.1.2: The status register is an 8-bit register separate from the other 32. Its eight bits are used as flags.

The microcontroller's pins are separated into 3 groups, B (digital), C (analogue), and D (digital).

Definition 8.2.1.3: The direction of each pin of a port is controlled by a data direction registers. We say a bit is set when it is 1 and the corresponding pin is output, and that a bit is cleared when it is 0 and the pin is input.

8.3 Programming in AVR Assembly

8.3.1 Directives

Definition 8.3.1.1: The assembly code contains statements called directives. These do not correspond to assembly instructions, but the assembler will take them into account when compiling. In AVR assembly, a directive is preceded by a dot. Some examples include:

- *.equ* assigns a value to a label which can be used in later expressions. This is a constant. (*.equ pi, 3*).
- *.global start* and *.global btnLED* declare the functions *start* and *btnLED* as global.

9 Databases

9.1 Database Design

9.1.1 The ANSI-SPARC Architecture

The architecture, set out by ANSI (the American National Standards Institute) and SPARC(the Standards, Planning, and Requirements Committee), consists of three levels used to realise data abstraction allowing different users to access the same data in different views without affecting each other. These levels are:

- **Definition 9.1.1.1:** The external layer is the user's view of the database. It can be adapted by the user to fit their specific needs without affecting others. This layer also incorporates derived, calculated, or combined fields.
- **Definition 9.1.1.2:** The conceptual layer is the community view of the database. It describes what data is stored in the database alongside the logical structure of the entire database.
- **Definition 9.1.1.3:** The internal layer is the physical representation of the database which describes how the data is stored to achieve optimal performance and storage utilisation.

Definition 9.1.1.4: A database schema is a complete description of the database as a result of the database design process. Each database should contain multiple external schemata, a conceptual schema, and an internal schema.

Definition 9.1.1.5: A database instance is the concrete data contained within the database at any particular time.

Definition 9.1.1.6: A database is considered to have logical data independence if the external schemata remain the same if the logical structure is changed on a conceptual level.

Definition 9.1.1.7: A database is considered to have physical data independence if the conceptual schema stays the same when the internal schema is changed.

9.1.2 Designing a Database

Definition 9.1.2.1: When creating a database, the first thing to do is to create a conceptual design. This is a high-level model of the data using entity-relationship modelling, identifying the appropriate entities, attributes, relationships, and constraints. This should be independent of any physical constraints.

Definition 9.1.2.2: The next step is to create a logical design which will be used to construct the relational data model using the conceptual design as a guide to transform entities and relationships into tables, using normalisation techniques to eliminate redundancies.

Definition 9.1.2.3: The physical design should be created last, and will describe how the logical design will be implemented, defining storage, access, and security details.

9.2 Modelling Databases

9.2.1 Entity-Relationship Modelling

Definition 9.2.1.1: An entity should be explicitly represented, denoted by a labelled rectangle and split into types and instances.

Definition 9.2.1.2: A relationship is a named association between two (or more) entity types which is represented by a labelled line with a triangle indicating reading direction.

Definition 9.2.1.3: Constraints are commonly represented on an entity-relationship model by a range ($min...max$). This can also include the cardinality (one-to-one, many-to-one, etc.), and participation (optional or mandatory).

Definition 9.2.1.4: The properties of an entity type are known as its attributes. In an entity-relationship model they are placed below an entities title with the primary key underlined.

9.2.2 Relational Data Models

Definition 9.2.2.1: A relational data model is based on the concept of mathematical relations and is the most widely used data structure for designing databases. It models a database by storing the relations between data in tables.

When creating a relational data model, the design must meet the following integrity constraints:

- **Definition 9.2.2.2:** Domain constraints for attributes limit the range of values permitted.
- **Definition 9.2.2.3:** Entity integrity ensures that the primary key is not null, guaranteeing that each entity has a unique identifier and that foreign keys form a valid reference.
- **Definition 9.2.2.4:** Referential integrity ensures that all foreign keys match some primary key or is null. This ensures that all references are valid.

9.2.3 Functional Dependencies

Definition 9.2.3.1: A relation's functional dependencies describe the relationship among its attributes.

Definition 9.2.3.2: If A and B are two sets of attributes in the same relation, where A is associated with exactly one value of B , then A functionally determines B and B is functionally dependent on A . A is called the determinant of B . Functional dependencies can take one of three forms:

- **Definition 9.2.3.3:** A full functional dependency exists when B is functionally dependent on A , but not on any proper subset of A .
- **Definition 9.2.3.4:** A partial functional dependency exists when B is functionally dependent on at least one proper subset of A .
- **Definition 9.2.3.5:** A transitive functional dependency exists between A and C if $A \rightarrow B$ and $B \rightarrow C$.

Definition 9.2.3.6: Given a set of functional dependencies F , its closure F^+ are all the functional dependencies which are implied by the dependencies in F . These can be calculated using Armstrong's Axioms.

Definition 9.2.3.7: Armstrong's Axioms are:

- **Definition 9.2.3.8:** Reflexivity: If $B \subseteq A$, then $A \rightarrow B$.
- **Definition 9.2.3.9:** Augmentation: If $A \rightarrow B$, then $(A, C) \rightarrow (B, C)$.
- **Definition 9.2.3.10:** Transitivity: If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$.

These rules are complete and sound, and can be used to derive another three rules:

- **Definition 9.2.3.11:** Decomposition: If $A \rightarrow (B, C)$, then $A \rightarrow B$ and $A \rightarrow C$.
- **Definition 9.2.3.12:** Union: If $A \rightarrow B$ and $A \rightarrow C$, then $A \rightarrow (B, C)$.
- **Definition 9.2.3.13:** Composition: If $A \rightarrow B$ and $C \rightarrow D$, then $(A, C) \rightarrow (B, D)$.

The closure F^+ of F can be computed by taking of F 's dependencies, repeatedly applying Armstrong's Axioms, and adding any new dependencies discovered in this way.

9.2.4 Normalisation

Definition 9.2.4.1: A table is in first normal form if it has no repeating groups (every cell has one value).

Definition 9.2.4.2: A table is in second normal form if it is in first normal form and has no partial dependencies (every non-key attribute is dependent on the whole primary key).

Definition 9.2.4.3: A table is in third normal form if it is in second normal form and has no transitive dependencies.

9.3 Relational Calculus and Algebra

9.3.1 Database Languages

Definition 9.3.1.1: Structured query language is the most common database language. It follows the ISO/ANSI standard using simple syntax with its two components, DDL and DML:

- **Definition 9.3.1.2:** Data definition language is used to define the schema for each relation, the domain for each respective attribute, and to specify the relation's integrity constraints.
- **Definition 9.3.1.3:** Data manipulation language is used to insert, update, delete, and retrieve data from the database.

Database languages can be separated into one of two types:

- **Definition 9.3.1.4:** A declarative language specifies which data needs to be retrieved in a query (relational calculus).
- **Definition 9.3.1.5:** A procedural language specifies how to retrieve the requested data for the query (relational algebra).

9.3.2 Relational Calculus

Definition 9.3.2.1: Relational calculus uses set theoretic expressions to define new sets based on one or more existing ones.

Definition 9.3.2.2: A predicate is a function that returns either true or false when given a set of arguments.

Definition 9.3.2.3: A proposition is the expression obtained when replacing values for the arguments of a predicate.

For example, for the predicates $P(x)$ and $Q(x)$, the set such that both $P(x)$ and $Q(x)$ are true is

$$\{x | P(x) \wedge Q(x)\} \quad (32)$$

9.3.3 Relational Algebra

Definition 9.3.3.1: Relational algebra uses operators to construct new sets from one or more existing ones.

Relational algebra uses a unique set of operations and symbols. These include:

- **Definition 9.3.3.2:** The selection operation, $\sigma_c(R)$, is a unary operation which selects the subset of tuples from a relation R that meet the specified condition c .
- **Definition 9.3.3.3:** The projection operation, $\Pi_{c_1, c_2}(R)$, is a unary operation which selects a specified subset of attributes c_1, c_2 from a relation R .
- **Definition 9.3.3.4:** The cartesian product operation, $A \times B$, is a binary operation which generates all possible combinations of tuples from A and B by concatenating them.
- **Definition 9.3.3.5:** The rename operation, $\rho_{Y(A, B, C)}(X)$, is a unary operation which takes the relation X and renames it to Y , and naming its corresponding attributes A, B, C .
- **Definition 9.3.3.6:** The division operation, $A \div B$, is a binary operation that only keeps tuples from A that exist in all possible combinations with B where B contains a subset of A .
- There are also join operations, corresponding to their SQL equivalents: inner $\bowtie$, left outer $\ltimes$, right outer $\rtimes$, and full outer $\Join$.
- Some other basic operations include $A - B$, $A \cup B$, and $A \cap B$.

10 Operating Systems

10.1 Introduction to Operating Systems

10.1.1 Basic Definitions

Definition 10.1.1.1: An operating system is a program that acts as an intermediary between a user and the hardware. Its goal is to execute programs, make solving problems easier, make the computer system convenient to use, and to use the resources of the system fairly and efficiently. The operating system is:

- **Definition 10.1.1.2:** A resource allocator. It is responsible for the management of the computer system's resources.
- **Definition 10.1.1.3:** A control program. It controls the execution of user programs and operation of the input/output devices to prevent errors and improper use of the computer.
- **Definition 10.1.1.4:** A kernel. It is the one program that runs all of the time. It consists of: the privileged instruction set, an interrupt mechanism, memory protection, and a real-time clock. It also has:

- **Definition 10.1.1.5:** A first-level interrupt handler which determines the source of an interrupt and initiates servicing the interrupt.
- **Definition 10.1.1.6:** The dispatcher which assigns processing resources for processes and is initiated when a current process cannot continue, or if the CPU may be used better elsewhere.
- Intra operating system communications.

10.1.2 Operating System Design

Definition 10.1.2.1: The operating system generally runs in dual-mode operation. This allows the operating system to protect itself and other system components. It differentiates which mode it is currently in using the mode bit.

- **Definition 10.1.2.2:** When the computer is in user mode it is limited as to which instructions it can execute, with some requiring switching modes to,
- **Definition 10.1.2.3:** Kernel mode (also known as system or privileged mode). In this mode, the system is given access to a set of privileged instructions which can perform tasks such as managing interrupts, performing inputs/outputs, and halting a process.

The operating system offers a set of services, some of which are:

- Providing an environment for the execution of programs and services.
- Providing a user interface, which can either be on the command line (CLI), graphical (GUI), or in Batch.
- Controlling the operation of input/output devices.
- Manipulating the file system.
- Organising communications either between processes on the same computer, or between computers over a network.
- Providing error detection, taking appropriate action to ensure correct and consistent computing.
- Allocating resources, particularly when multiple jobs are running concurrently.
- Providing protection and security by ensuring that all access to system resources are controlled and that all users are authenticated.

10.1.3 Processes and Threads

Definition 10.1.3.1: A program in execution is known as a process. A process includes code, its current activity, a stack, a data section, and a heap.

Definition 10.1.3.2: A process's information is stored in a process control block. These PCBs are then arranged together into a process table. Each PCB stores a process's unique identifier, state, memory usage, CPU scheduling information, and its corresponding CPU registers.

Definition 10.1.3.3: As a process is executed, it changes state. Here, it can take one of five states: new, when the process is being created; running, when instructions are being executed; waiting, when the process is waiting for some event to occur; ready, when the process is ready to be dispatched to the CPU; and terminated, when the process has finished executing.

Definition 10.1.3.4: A process can, as a parent process, create a number of child processes, which in turn can create other processes, forming a tree of processes. A parent process may share some of its resources with its children, and they can either execute concurrently or sequentially.

Definition 10.1.3.5: The process scheduler selects which of the currently available processes will be executed next by the CPU. It maintains the following scheduling queues:

- **Definition 10.1.3.6:** The job queue is the set of all processes in the system.
- **Definition 10.1.3.7:** The ready queue is the set of all processes residing in main memory, ready and waiting to execute.
- **Definition 10.1.3.8:** The device queues are sets of processes waiting for I/O devices.

Definition 10.1.3.9: A thread is a unit of execution which can be traced and that belongs to a process. It can either be a user or kernel thread.

The thread model of processing uses local variables within threads, with global variables being available to all threads. The model is frequently used as: it requires less time to create or terminate threads vs. processes; it may allow for execution to continue, even if part of a process is blocked; threads can share processing resources; creating threads is cheaper than creating processes or context switching; and processes can take advantage of multiprocessor architectures, increasing scalability.

Definition 10.1.3.10: Amdah's Law states that performance gains from adding additional cores with serial and parallel components follows the following equation:

$$speedup \leq \frac{1}{S + \frac{(1-S)}{N}} \quad (33)$$

where S is the serial portion and N is the number of processing cores.

For example, if an application is 75% parallel, moving from one to two cores will result in a speedup of 1.6 times. As N approaches infinity, the speedup approaches $\frac{1}{S}$.

Multithreading can be accomplished by using one of three models:

- **Definition 10.1.3.11:** The many-to-one model maps many user-level threads to a single kernel thread.
- **Definition 10.1.3.12:** The one-to-one model maps each user-level thread to a kernel thread, allowing for greater concurrency.
- **Definition 10.1.3.13:** The many-to-many model allows many user-level threads to be mapped to many kernel threads.

10.2 Process, Memory, and Storage Management

10.2.1 CPU Scheduling

Definition 10.2.1.1: The aim of multiprogramming is to maximise CPU utilisation by keeping multiple processes in memory concurrently, meaning the CPU can always be executing a ready process whilst another is waiting.

Definition 10.2.1.2: The long-term (job) scheduler selects which processes should be brought into the ready queue.

Definition 10.2.1.3: The medium-term scheduler is a part of swapping and is in-charge of handling the swapped out processes.

Definition 10.2.1.4: The short-term (CPU) scheduler selects the process to be executed next and allocates it to the CPU.

Definition 10.2.1.5: The dispatcher is the module that gives control of the CPU to the process selected by the CPU scheduler. It is in charge of context switching, switching to user mode, and jumping to the proper location in the user program to restart that program.

Definition 10.2.1.6: The dispatch latency is the time it takes for the dispatcher to stop one process and start another running.

When scheduling processes, the criteria considered are:

- CPU utilisation.
- **Definition 10.2.1.7:** Throughput, the number of processes that complete their execution per unit of time.
- **Definition 10.2.1.8:** Turnaround time, the amount of time to execute a particular process (waiting, execute, and I/O).
- **Definition 10.2.1.9:** Waiting time, the total amount of time a process has been waiting in the ready queue.
- **Definition 10.2.1.10:** Response time, the amount of time it takes from when a request was submitted until the first response.

This can be achieved using one of the following algorithms: first come, first served; shortest job first; shortest remaining time first; priority; round-robin; or multilevel/feedback queue.

10.2.2 Memory Management

Definition 10.2.2.1: Each process has a separate memory space. A pair of base and limit registers define the logical address space. The CPU must check every memory access generated in user mode to be sure it is between base and limit for that user.

Definition 10.2.2.2: The logical address space is generated by the CPU and can also be referred to as the virtual address.

Definition 10.2.2.3: The physical address space is the address seen by the memory unit. This address cannot be seen by user applications.

Definition 10.2.2.4: The memory management unit is a hardware device that maps virtual addresses to their corresponding physical ones.

Definition 10.2.2.5: Relocation registers are used to protect user processes from each other, and from changing operating-system code and data. This is known as memory protection.

Memory can be allocated into one of a set of holes using either: first-fit, putting the memory in the first hole that is big enough; best-fit, putting the memory in the smallest hole which is big enough; and worst-fit, putting the memory into the largest available hole.

Definition 10.2.2.6: External fragmentation is memory space that is able to satisfy a request but it is not contiguous.

Definition 10.2.2.7: Internal fragmentation is memory that is allocated successfully, but may be slightly larger than the requested amount of memory.

Definition 10.2.2.8: External fragmentation can be reduced by compaction, where memory contents are shuffled to place all free memory together in one large block.

Definition 10.2.2.9: Physical memory is divided into fixed-sized blocks called frame while logical memory is divided into blocks of the same size called pages.

Definition 10.2.2.10: To run a program of size n pages, we need to find n free pages before loading the program. This process is called paging.

Definition 10.2.2.11: A page table is set up to help manage the mapping of logical to physical addresses using an address translation.

Definition 10.2.2.12: The logical memory address generated by the CPU is divided into the page number, which is used as an index into the page table which contains the base address of each page in physical memory, and the page offset, which combined with the base address to define the physical memory address that is sent to the memory unit.

Definition 10.2.2.13: When a logical address is generated by the CPU, the MMU first checks if its page number is present in the TLB. If the page number is found, its frame number is immediately available and is used to access memory. If the page number is not in the TLB, a memory reference to the page table has to be created. If the TLB is already full of entries, then an existing entry is selected for replacement.

Definition 10.2.2.14: The hit ratio α , is the percentage of times that a page number is found in the associative registers.

Definition 10.2.2.15: The effective access time is calculated using

$$\text{EAT} = (\alpha \times \text{memory access time}) + 2((1 - \alpha) \times \text{memory access time}) \quad (34)$$

Definition 10.2.2.16: Memory protection is implemented by associating a protection bit with each entry. This is done by attaching a valid-invalid bit to each entry where valid indicates that the associated page is in the process's logical address space, and thus is a legal page.

Definition 10.2.2.17: The page fault rate is the percentage of page references which are invalid (represented as a decimal).

Definition 10.2.2.18: Demand paging limits the pages in memory to those which are currently required by the process during execution. This reduces the number of pages to be processed and the number of frames allocated to a process whilst increasing overall response time for a collection of processes and the number of processes executing.

Definition 10.2.2.19: Page replacement finds some page in memory that isn't really in use and swaps it out for some other required page.

Definition 10.2.2.20: Belady's anomaly states that, for some page fault algorithms, the page fault rate may increase as the number of allocated frames increases.

Definition 10.2.2.21: Thrashing occurs when a process is spending more time paging than doing actual work. If a process does not have enough pages, then the page-fault rate is very high.

Definition 10.2.2.22: Prepaging is the process of moving all the required pages into memory at once.

10.2.3 Storage Management

The performance of a disk storage device is dependent on four factors: seek time, the time to move the head to the target track; rotational delay/latency, the time waiting for the target sector to rotate under the head; access time, seek time + latency; and transfer time.

Definition 10.2.3.1: Disk bandwidth is the total number of bytes transferred, divided by the total time between the first request for service and the completion of the last transfer.

Definition 10.2.3.2: Network-attached storage is storage made available over a network rather than over a local connection.

Definition 10.2.3.3: Redundant array of inexpensive disk uses multiple disk drives to increase reliability via redundancy. It is arranged in seven levels: non-redundant striping, mirrored disks, memory-style error-correcting codes, bit-interleaved parity, block-interleaved parity, block-interleaved distributed parity, P+Q redundancy.

10.2.4 Deadlocks

Definition 10.2.4.1: A set of processes is deadlocked if each process in the set is waiting for an event that only another process in the set can cause. A deadlock must fulfill the following conditions:

- **Definition 10.2.4.2:** Mutual exclusion, only one process at a time can use a resource.
- **Definition 10.2.4.3:** Hold and wait, a process holding at least one resource is waiting to acquire additional resources held by other processes.
- **Definition 10.2.4.4:** No preemption, a resource can be only be released voluntarily by the process holding it, after that a process has completed its task.
- **Definition 10.2.4.5:** Circular wait, there exists a set of waiting processes such that the first process is waiting for a resource from the second process, which is waiting for a resource from the third process, and so on such that the final process is waiting on a resource from the first process.

Definition 10.2.4.6: A resource-allocation graph has a set of vertices separated into two sets, processes and resources, and a set of edges, where each of which can either be a request or assignment edge.

We can deal with deadlocks in one of the following ways:

- **Definition 10.2.4.7:** The ostrich algorithm where we simply pretend there is no problem.
- **Definition 10.2.4.8:** Deadlock prevention where processes are limited to holding one resource at a time.
- **Definition 10.2.4.9:** Deadlock avoidance where each process declares the maximum number of resources of each type that it might need.
- **Definition 10.2.4.10:** Process termination is used to recover from deadlocks by selecting a victim with the minimum cost, or rolling back to a safe state.

Definition 10.2.4.11: The system is in a safe state if there exists a sequence of all the processes in the system such that, for each process, the resources it is able to request can be satisfied from currently available resources.

Part IV

Mathematics for Computer Science

11 Calculus

11.1 Introduction to Calculus

11.1.1 Sets

Definition 11.1.1.1: A set is an unordered collection of distinct elements denoted as $A = \{x | \text{condition}\}$.

Some common set operations include:

- **Definition 11.1.1.2:** The union of two sets, $A \cup B$, includes all the elements of both sets.
- **Definition 11.1.1.3:** The intersection of two sets, $A \cap B$, includes any elements which are present in both sets.
- **Definition 11.1.1.4:** If A and B are finite sets, then the principle of inclusion-exclusion states that

$$A \cup B = A + B - (A \cap B) \quad (35)$$

- **Definition 11.1.1.5:** The set A is a subset of B , $A \subseteq B$, if every element of A is an element of B . A is a proper subset of B , $A \subset B$, if A is a subset of B and there is at least one element of B not in A .
- **Definition 11.1.1.6:** The powerset of a set A is the set of all subsets of A .

$$\mathcal{P}(A) = \{A' | A' \subseteq A\} \quad (36)$$

- **Definition 11.1.1.7:** The cartesian product of two sets, $A \times B$, is the set of all possible ordered pairs with the first element from A and the second from B .

Definition 11.1.1.8: For a finite set, the cardinality is the number of distinct elements that it contains.

11.1.2 Functions

For a function $f : A \rightarrow B$,

- **Definition 11.1.2.1:** A is known as the domain of f .
- **Definition 11.1.2.2:** B is known as the codomain of f .
- **Definition 11.1.2.3:** If $f(a) = b$, then b is the image of a .
- **Definition 11.1.2.4:** The pre-image of b is the set $\{a | f(a) = b\}$.
- **Definition 11.1.2.5:** f forms an injection if every element of A maps to a unique element in B , even if some elements of B don't have a mapping.
- **Definition 11.1.2.6:** f forms a surjection if every element of B maps to at least one element of A .
- **Definition 11.1.2.7:** f forms a bijection if it is both injective and surjective.

Definition 11.1.2.8: If for an unbounded set A , there is a bijective function $f : A \rightarrow \mathbb{N}$, then we say A is countably infinite, otherwise A is known as uncountably infinite.

11.1.3 Hyperbolic Functions

Definition 11.1.3.1: Given the unit hyperbola $x^2 - y^2 = 1$, the hyperbolic functions are defined by the coordinates of a point at which area $\frac{1}{2}\alpha$ has been swept out.

$$\begin{aligned} \sinh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{2} & \cosh(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{2} & \tanh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{e^\alpha + e^{-\alpha}} & \sigma(\alpha) &= \frac{1}{1 + e^{-\alpha}} \\ \operatorname{sech}(\alpha) &= \frac{2}{e^\alpha + e^{-\alpha}} & \operatorname{cosech}(\alpha) &= \frac{2}{e^\alpha - e^{-\alpha}} & \coth(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{e^\alpha - e^{-\alpha}} \end{aligned}$$

These functions can then be differentiated as follows:

$$\begin{aligned} \frac{d}{dx} [\sinh(\alpha)] &= \cosh(\alpha) & \frac{d}{dx} [\cosh(\alpha)] &= \sinh(\alpha) \\ \frac{d}{dx} [\tanh(\alpha)] &= \operatorname{sech}^2(\alpha) & \frac{d}{dx} [\sigma(\alpha)] &= \sigma(\alpha)(1 - \sigma(\alpha)) \end{aligned}$$

11.2 Multivariate Functions

11.2.1 Multivariate Derivatives

Definition 11.2.1.1: For a function $f(x, y)$, if we take $f = f_0$ to be fixed but unknown, then we can differentiate f as a function of x at $x = x_0$, giving us the partial derivative of f with respect to x which gives the gradient when moving in the x -direction from any point.

$$\frac{\delta f}{\delta x} [f(x, y)] = \lim_{x \rightarrow x_0} \left(\frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \right) = f_x \quad (37)$$

Definition 11.2.1.2: The vector of partial derivatives, denoted as ∇f , contains all the partial derivatives of a function. This forms a vector pointing in the direction of the greatest rate of increase of f , where its magnitude is the rate of increase in that direction. For example, for a function $f(x, y)$,

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (38)$$

Definition 11.2.1.3: The directional derivative of the function $f(x, y)$ gives the gradient at any point whilst looking in the direction given by the unit vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$.

$$\nabla_{\mathbf{v}} f = \lim_{h \rightarrow 0} \frac{f(x_0 + h\mathbf{v}) - f(x_0)}{h} \quad (39)$$

If the slope is approximately flat at a small scale, then the slope can be determined by

$$\nabla_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f \quad (40)$$

From this, we get that, for a function $f(\mathbf{x}_0)$ and a direction $\mathbf{v}$,

$$f(\mathbf{x}_0 + h\mathbf{v}) \approx f(\mathbf{x}_0) + h\mathbf{v} \cdot \nabla f(\mathbf{x}_0) \quad (41)$$

11.2.2 Multivariate Extrema

Definition 11.2.2.1: A function that is continuous is in class C^0 , and a function that is continuous and has a continuous derivative is in class C^1 . A function f such that f' , f'' , ..., $f^{(k)}$ exist and are continuous is in class C^k .

Definition 11.2.2.2: A function f such that derivatives of all orders exist and are continuous is in class C^∞ is also smooth.

Definition 11.2.2.3: All the second-order partial derivatives of a function $f(x, y)$ can be gathered into a Hessian matrix.

$$\mathbf{H}_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \quad (42)$$

This can be used to determine the form of bivariate extrema. Suppose $f(x, y) \in C^2$ and $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$ then,

- (x_0, y_0) is a local maximum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} < 0$ at (x_0, y_0) .
- (x_0, y_0) is a local minimum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} > 0$ at (x_0, y_0) .
- (x_0, y_0) is a saddle point if $\det(\mathbf{H}_f) < 0$.
- If $\det(\mathbf{H}_f) = 0$, then the test is inconclusive.

At a stationary point x_0 , the quadratic approximation of a function $f(x)$ is

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} \mathbf{v}^T \mathbf{H}_f(x_0) \mathbf{v} \quad (43)$$

But as $\mathbf{H}_f(x_0)$ is real and symmetric, it has orthogonal eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, so we can write

$$\mathbf{v} = c_1 \mathbf{e}_1 + c_2 \mathbf{e}_2 + \dots + c_n \mathbf{e}_n \quad (44)$$

When substituted back into the original equation, this gives that

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} (\lambda_1(c_1)^2 + \lambda_2(c_2)^2 + \dots + \lambda_n(c_n)^2) \quad (45)$$

Given this, we can use the eigenvalues to determine the nature of the stationary point x_0 ,

- If all eigenvalues are positive, then the stationary point is a local minimum.
- If all eigenvalues are negative, then the stationary point is a local maximum.
- If there is a mixture of positive and negative eigenvalues, then the stationary point is a saddle point.

11.2.3 Vector-Valued Functions

Definition 11.2.3.1: A vector-valued function takes an n -dimensional vector as input, and outputs an m -dimensional vector, e.g., $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Definition 11.2.3.2: The Jacobian matrix of a vector-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be formed by setting $\mathbf{J}_{i,j} = \frac{\delta f_i}{\delta x_j}$ to form,

$$\mathbf{J}_f = \begin{pmatrix} \frac{\delta f_1}{\delta x_1} & \dots & \frac{\delta f_1}{\delta x_n} \\ \vdots & \ddots & \vdots \\ \frac{\delta f_m}{\delta x_1} & \dots & \frac{\delta f_m}{\delta x_n} \end{pmatrix} \quad (46)$$

For a multivariate function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian matrix can be used to linearly approximate f at x ,

$$f(x + h\mathbf{v}) \approx f(x) + \mathbf{J}_f(x)(h\mathbf{v}) \quad (47)$$

11.3 Lagrange Multipliers

11.3.1 Constrained Optimisation

Definition 11.3.1.1: The goal of a constrained optimisation problem is to either minimise or maximise a multivariate function $f(\mathbf{x})$ where $x_1, x_2, \dots, x_n$ are not independent subject to some constraint $g(\mathbf{x})$.

Definition 11.3.1.2: The Lagrange multipliers method is used to find stationary points of an objective function $f(\mathbf{x}) \in C^1$ subject to the constraint $g(\mathbf{x}) = 0$.

1. Define a new objective function called the Lagrangian,

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda g(\mathbf{x}) \quad (48)$$

2. Find the stationary points of $\mathcal{L}$ with respect to both $\mathbf{x}$ and λ , i.e., solve the system of equations $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$ and $g(\mathbf{x}) = 0$ for the unknowns $\mathbf{x}$ and λ .
3. Then, the constrained extrema are found among the solutions to these equations. Identify their nature.

11.4 Series

11.4.1 Introduction to Series

Definition 11.4.1.1: A sequence is an ordered, countably infinite, set of numbers.

Definition 11.4.1.2: A series is the sum of the members of a sequence.

Theorem 11.4.1.1: If the series $\sum_{n=0}^{\infty} a_n$ and the series $\sum_{n=0}^{\infty} b_n$ are convergent, and have respective sums α and β , then

$$\sum_{n=0}^{\infty} (a_n + b_n) = \alpha + \beta \quad (49)$$

and

$$\sum_{n=0}^{\infty} (a_n - b_n) = \alpha - \beta \quad (50)$$

Proof: $\sum_{n=0}^{\infty} a_n$ converges to α and $\sum_{n=0}^{\infty} b_n$ converges to β . Given an arbitrary $\epsilon > 0$, let $\epsilon_a = \frac{1}{2}\epsilon = \epsilon_b$. Since $\sum_{n=0}^{\infty} a_n$ converges, $\exists N_a$ such that,

$$\left| \sum_{r=0}^n a_r - \alpha \right| < \epsilon_a, \quad \forall n > N_a \quad (51)$$

Similarly, since $\sum_{n=0}^{\infty} b_n$ converges, $\exists N_b$ such that,

$$\left| \sum_{r=0}^n b_r - \beta \right| < \epsilon_b, \quad \forall n > N_b \quad (52)$$

Taking $N = \max\{N_a, N_b\}$, then $\forall n > N$,

$$\begin{aligned} \left| \sum_{r=0}^n (a_r + b_r) - (\alpha + \beta) \right| &= \left| \left(\sum_{r=0}^n a_r - \alpha \right) + \left(\sum_{r=0}^n b_r - \beta \right) \right| \\ &\leq \left| \sum_{r=0}^n a_r - \alpha \right| + \left| \sum_{r=0}^n b_r - \beta \right| \leq \epsilon_a + \epsilon_b = \epsilon \end{aligned} \quad (53)$$

i.e, $\sum_{n=0}^{\infty} (a_n + b_n)$ converges to $(\alpha + \beta)$ using triangle inequality $|a + b| \leq |a| + |b|$.

□

Theorem 11.4.1.2: If $\lim_{n \rightarrow \infty} a_n \neq 0$ then the series $\sum_{n=0}^{\infty} a_n$ diverges.

Proof: Suppose $\sum_{n=0}^{\infty} a_n$ converges for some S , Then, $\lim_{n \rightarrow \infty} S_n = S$, which means that given arbitrary $\epsilon > 0$, there exists some N_ϵ such that, for all $n > N_\epsilon$,

$$|S_n - S| < \epsilon \quad |S_n - S| < 2\epsilon \quad (54)$$

Combining these,

$$2\epsilon > |S_n - S| + |S_{n+1} - S| = |S - S_n| + |S_{n+1} - S| \geq |S_{n+1} - S_n| = |a_{n+1}| \quad (55)$$

But ϵ was arbitrary, so we have shown that, for any $\epsilon' > 0$ there exists an N_ϵ such that $|a_n| < \epsilon'$ for all $n > N_\epsilon$.

□

As the number of elements in a series approaches infinity...

- **Definition 11.4.1.3:** If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series diverges.
- **Definition 11.4.1.4:** If $\lim_{n \rightarrow \infty} a_n = 0$, then the series converges.
- **Definition 11.4.1.5:** If $\lim_{n \rightarrow \infty} |a_n| = 0$, then the series absolutely converges.
- **Definition 11.4.1.6:** If $\lim_{n \rightarrow \infty} a_n = S$, then the series is convergent to S .

Theorem 11.4.1.3: The ratio test states that, if $\sum_{n=1}^{\infty} a_n$ is a series where $a_n \neq 0$ and $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ then

- a) If $L < 1$, the series absolutely converges.
- b) If $L > 1$, the series is divergent.
- c) If $L = 1$, then the test fails.

Proof: (a) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$, then $\exists r$ with $L < r < 1$ and N such that $\forall n > N$:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < r \quad \rightarrow \quad |a_{n+1}| < r|a_n| \quad (56)$$

Then $|a_{n+2}| < r|a_{n+1}| < r^2|a_n|$ and $|a_{n+3}| < r|a_{n+2}| < r^3|a_n|$ etc. So the remainder is

$$R_{n+m} = \lim_{m \rightarrow \infty} \left(|a_n| r^{m+1} \frac{1}{1-r} \right) = 0 \quad (57)$$

Hence the series converges. □

Definition 11.4.1.7: The n^{th} root test states that, for the series $\sum_{n=1}^{\infty} a_n$, $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$,

- If $L < 1$, then the series absolutely converges.
- If $L > 1$, then the series is divergent.
- If $L = 1$, then the test fails.

Definition 11.4.1.8: The alternating series test states that the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges if $a_n > 0$, $a_{n+1} \leq a_n \forall n$, and $\lim_{n \rightarrow \infty} a_n = 0$

Theorem 11.4.1.4: Every subsequence of a convergent sequence converges to the same limit as the original sequence.

Proof: Let $\{a_{n_k}\}$ be a subsequence of $\{a_n\}$. First note that $n_k \geq k$ and so $n_{k+1} \geq k+1$. Let $\lim_{n \rightarrow \infty} a_n = L$. Then, for any $\epsilon > 0$, there exists N_ϵ such that for all $n > N_\epsilon$ it holds that $|a_n - L| < \epsilon$. Let $k > N_\epsilon$, then $n_k > N_\epsilon$ and so it must be that $|a_{n_k} - L| < \epsilon$. □

11.4.2 Power Series

Definition 11.4.2.1: A power series is a series involving a variable x and a constant x_0 of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots \quad (58)$$

At any specific value of x , the power series becomes a normal series.

Definition 11.4.2.2: For an arbitrary power series $\sum_{n=0}^{\infty} a_n (x - x_0)^n$, we can use the ratio test (including the $(x - x_0)$ term). From this, we call $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = r$ the radius of convergence. If $|x - x_0| < r$, we have absolute convergence and if $|x - x_0| > r$, we have divergence.

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $r > 0$. Then, within the interval $(-r, r)$:

- $f(x)$ is continuous.
- $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ (term-by-term differentiation).
- $\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$ (term-by-term integration).
- Both series $f'(x)$ and $\int_0^x f(t) dt$ have the same radius of convergence as $f(x)$.

Definition 11.4.2.3: Suppose we start with some function $f(x)$ and we wish to determine a power series $\sum_{n=0}^{\infty} a_n x^n$ such that $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Putting these values in, we see that if such a power series exists, then:

$$f(x) = f(0) + f'(0)x + f''(0)\frac{x^2}{2!} + f^{(3)}(0)\frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} f^{(n)}(0)\frac{x^n}{n!} \quad (59)$$

This is called the Maclaurin series for f . To use the series, we must have that $f \in C^\infty$ and the power series converges.

Definition 11.4.2.4: We can recenter the power series on x_0 by defining a new function $g(x) = f(x + x_0)$ then $g(0) = f(x_0)$ and $g'(x) = f'(x + x_0)$. Repeatedly differentiating, $g^{(n)}(x) = f^{(n)}(x + x_0)$ so we build the Maclaurin series for g , which gives us

$$f(x + x_0) = g(x) = \sum_{n=0}^{\infty} g^{(n)}(0) \frac{x^n}{n!} = \sum_{n=0}^{\infty} f^{(n)}(x_0) \frac{x^n}{n!} \quad (60)$$

or,

$$f(x) = g(x - x_0) = f(x_0) + f'(x_0)(x - x_0) + f''(x_0) \frac{(x - x_0)^2}{2!} + \dots \quad (61)$$

This form is called the Taylor series expansion of f .

11.4.3 Taylor's Theorem

Lemma 11.4.3.1: If a real-valued function f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.

Proof: Not proved here.

□

Theorem 11.4.3.1: Suppose a function f is n times differentiable on $[a, x]$, then there exists some $\xi \in (a, x)$ such that

$$f(x) = f(a) + (x - a)f'(a) + \dots + \frac{(x - a)^{(n-1)}}{(n - 1)!} f^{(n-1)}(a) + R^n(x) \quad (62)$$

Where the remainder term is equal to $R_n(x) = \frac{(x-a)^n}{n!} f^{(n)}(\xi)$.

Proof: Suppose a function f is n times differentiable on $[a, x]$. Let k be a constant defined so that

$$f(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!} f''(a) + \dots + \frac{(x - a)^{(n-1)}}{(n - 1)!} f^{(n-1)}(a) + \frac{(x - a)^n}{n!} k \quad (63)$$

We can then define the function

$$F(y) = f(x) - f(y) - (x - y)f'(y) - \dots - \frac{(x - y)^{(n-1)}}{(n - 1)!} f^{(n-1)}(y) - \frac{(x - y)^n}{n!} k \quad (64)$$

Then $F(x) = 0$ and also $F(a) = f(x) - f(x) = 0$ so we can apply Rolle's theorem to F on $[a, x]$, meaning there is some $\xi \in (a, x)$ such that $F'(\xi) = 0$.

□

11.4.4 Fourier Series

Definition 11.4.4.1: For a function $f(x)$, $-\pi < x < \pi$, the Fourier series is

$$f(x) \approx \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \quad (65)$$

where

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \quad (66)$$

12 Linear Algebra

12.1 Introduction to Linear Algebra

12.1.1 Mathematical Structures

Definition 12.1.1.1: Abstraction is the process of simplifying a complex problem via removing unnecessary details in order to apply more general problem solving techniques.

Definition 12.1.1.2: Groups consist of a set G and a binary operation $*$. They abide by the following axioms:

- **Definition 12.1.1.3:** Closure, when applying an operation between two elements of the group, the result will also be an element of the group.

$$\forall a, b \in G, a * b = c, c \in G \quad (67)$$

- **Definition 12.1.1.4:** Associativity, for any elements a, b, c in the group, the order in which the operation is performed does not matter, as long as the elements stay in their original order.

$$\forall a, b, c \in G, a * (b * c) = (a * b) * c \quad (68)$$

- **Definition 12.1.1.5:** Identity, the group contains an element such that when an operation is performed between an element in the group and the identity, the original element is returned.

$$\exists e \in G, \forall a \in G, a * e = e * a = a \quad (69)$$

- **Definition 12.1.1.6:** Inverse, for every element in the group, there exists an inverse element such that when an operation is performed between them, the identity is returned.

$$\forall a \in G, \exists b \in G, a * b = e \quad (70)$$

Definition 12.1.1.7: Fields contain a set F with two binary operations (addition and multiplication) and abide by the following axioms:

- Associativity (defined above).
- **Definition 12.1.1.8:** Commutativity - when applying an operation between two elements of the field, the order the elements appear in does not matter.

$$\forall a, b \in F, a + b = b + a \text{ and } a \times b = b \times a \quad (71)$$

- **Definition 12.1.1.9:** Identity, there exists two distinct elements, 0 and 1, in F such that,

$$\exists 0, 1 \in F, \forall a \in F, a + 0 = a \text{ and } a \times 1 = a \quad (72)$$

- **Definition 12.1.1.10:** Additive inverses, for every element in the field, there exists some other element, called the additive inverse, such that,

$$\forall a \in F, \exists b \in F, a + b = 0 \quad (73)$$

- **Definition 12.1.1.11:** Multiplicative inverses, for every element in the field, there exists some other element, called the multiplicative inverse, such that,

$$\forall a \in F, \exists b \in F, a \times b = 1 \quad (74)$$

- **Definition 12.1.1.12:** Distributivity of multiplication over addition. This axiom states that,

$$\forall a, b, c \in F, a \times (b + c) = (a \times b) + (a \times c) \quad (75)$$

Definition 12.1.1.13: A vector space is a mathematical structure with two sets, scalars F and vectors V , where V forms a commutative group under addition and F forms a field. They abide by the following axioms:

- Associativity (defined above).
- Identity (defined above).
- Distributivity of multiplication over addition (defined above).

12.1.2 Linear Combinations and Maps

Definition 12.1.2.1: Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ and scalars $a_1, \dots, a_n$, we can form a linear combination.

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n = \sum_{i=1}^n a_i\mathbf{v}_i \quad (76)$$

Definition 12.1.2.2: The set of all linear combinations of a set of vectors is called its span, denoted $span(V)$.

Definition 12.1.2.3: A set of vectors S is a spanning set for a vector space V if $span(S) = V$. We can equivalently say that S spans V .

Definition 12.1.2.4: A set of vectors S is linearly dependent if there exists a linear combination such that

$$\sum_{i=1}^n a_i\mathbf{s}_i = \mathbf{0} \text{ and } \exists i, a_i \neq 0 \quad (77)$$

Otherwise, the set is described as linearly independent.

Definition 12.1.2.5: A subset S of the vector space V is a basis for V if S is a spanning set and is linearly independent.

Theorem 12.1.2.1: For a vector space V , if S is a basis for V , then it must also be a minimal spanning set.

Proof: Let S be a basis of V . Assume that S is not a minimal spanning set, then there must be some element $s \in S$ such that $span(S \setminus \{s\}) = V$. Then, s can be written as a linear combination of elements from $S \setminus \{s\}$. That is, we have that

$$1 \cdot \mathbf{s} - (a_1\mathbf{s}_1 + a_2\mathbf{s}_2 + \dots + a_n\mathbf{s}_n) = \mathbf{0} \quad (78)$$

This means that S would be linearly dependent, contradicting the definition of S as a basis. Therefore, S must be a minimal spanning set.

□

Definition 12.1.2.6: The dimension of a vector space V is the cardinality of its basis S .

Definition 12.1.2.7: If we have a function $f : V \rightarrow W$ where V and W are vector spaces over a field F , then f is a linear map if,

$$\exists \mathbf{u}, \mathbf{v} \in V, \exists a \in F, f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v}) \text{ and } f(a \cdot \mathbf{v}) = a \cdot f(\mathbf{v}) \quad (79)$$

These can also be called linear transformations or vector space homomorphisms.

Definition 12.1.2.8: If $V = W$ for a linear map f , then f is also a endomorphism.

12.2 Matrices

12.2.1 Matrix Operations

When the determinant of a matrix is zero, its columns are linearly dependent vectors. Therefore, if two columns are the same, then the determinant will be zero. Similarly, if a matrix A is identical to another matrix B , except for two columns being swapped, then $\det(A) = -\det(B)$.

Definition 12.2.1.1: To calculate a determinant, we use the Laplace expansion,

$$\det(C) = \sum_{j=1}^n (-1)^{i+j} c_{(i,j)} M_{(i,j)} \quad 0 < i \leq n, i \in \mathbb{N} \quad (80)$$

where $c_{(i,j)}$ is the coefficient of the term at (i,j) and $M_{(i,j)}$ is the minor at coordinates (i,j) .

Definition 12.2.1.2: Under a linear map $f : X \rightarrow Y$, $f(a\mathbf{x} + b\mathbf{y}) = af(\mathbf{x}) + bf(\mathbf{y})$, $f(X)$ is known as the column space and is the span of the columns. Its dimension is known as the column rank (this is also the number of linearly independent columns). Row space and rank can be defined similarly.

Definition 12.2.1.3: The rank of a matrix is its row/column rank after the matrix is put into RREF.

Definition 12.2.1.4: If $f : X \rightarrow Y$ is a linear map, then the kernel or null space of f is the set of vectors that map to zero. The dimension of the kernel is known as the nullity. The kernel forms a vector space where all linear combinations of kernel elements are in the kernel.

Definition 12.2.1.5: Given a linear map on vector spaces $T : V \rightarrow W$, if V has finite dimensions then $\text{Rank}(T) + \text{Nullity}(T) = \dim(V)$.

12.2.2 Systems of Equations and Gaussian Elimination

Definition 12.2.2.1: A finite set of linear equations is called a system of equations or a linear system.

- **Definition 12.2.2.2:** A solution is a sequence of numbers $s_1, s_2, \dots, s_n$ such that the assignment $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ satisfies every equation.
- **Definition 12.2.2.3:** A system is consistent if it has at least one solution.
- **Definition 12.2.2.4:** A system $A\mathbf{x} = \mathbf{b}$ is homogenous if $\mathbf{b}$ is a column of zeros. These systems have a trivial solution of $\forall x \in \mathbf{x}, x = 0$.

Definition 12.2.2.5: An augmented matrix is in row echelon form if the first non-zero number in each row is a one, any rows containing only zeros are grouped at the bottom, and if two successive rows are not all zeroes, then the leading one of the higher row is to the left of the leading one in the lower row.

Definition 12.2.2.6: An augmented matrix is in reduced row echelon form if it is in row echelon form and each column containing a leading one has zeroes everywhere else.

12.2.3 Geometry

Definition 12.2.3.1: Given a vector space X , a function $d : X \rightarrow \mathbb{R}$ is a norm if it satisfies:

- **Definition 12.2.3.2:** Triangular Inequality: $\forall x, y \in X, d(x + y) \leq d(x) + d(y)$.
- **Definition 12.2.3.3:** Absolute Homogeneity: $\forall x \in X, s \in \mathbb{R}, d(s \cdot x) = |s| \cdot d(x)$.
- **Definition 12.2.3.4:** Positive Definite: If $d(x) = 0$ then $x = 0$.

Definition 12.2.3.5: The Euclidean norm is the length of a vector and is given for $\mathbf{x} \in \mathbb{R}^n$ to be

$$\|\mathbf{x}\| = \sqrt{\sum_{i=1}^n (x_i)^2} \quad (81)$$

Any non-unit vector $\mathbf{x}$ can be used to define a unit vector $\hat{\mathbf{x}}$ in the direction of $\mathbf{x}$ by using

$$\hat{\mathbf{x}} = \frac{1}{\|\mathbf{x}\|} \mathbf{x} \quad (82)$$

Definition 12.2.3.6: The Manhattan distance is given by

$$d_M(\mathbf{x}) = \sum_{i=1}^n |x_i| \quad (83)$$

Definition 12.2.3.7: Given $p \in \mathbb{R}^+$, we can define the p -norm $\|\mathbf{x}\|_p$ on $\mathbb{R}^n$ as

$$\|\mathbf{x}\|_p = \sqrt[p]{\sum_{i=1}^n |x_i|^p} \quad (84)$$

Using limits on the p -norm, we get

$$\lim_{p \rightarrow \infty} \|\mathbf{x}\|_p = \max(\mathbf{x}) \quad \lim_{p \rightarrow 0} \|\mathbf{x}\|_p = \infty \quad (85)$$

Definition 12.2.3.8: Two vectors are orthogonal if their dot product is zero.

Definition 12.2.3.9: Two vectors are normal if their Manhattan norm is one.

Definition 12.2.3.10: A set of vectors is orthonormal if they are all both orthogonal and normal.

12.3 Eigenvectors and Eigenvalues

12.3.1 Introduction to Eigenvectors and Eigenvalues

Definition 12.3.1.1: Let A be an $n \times n$ matrix. A non-zero vector $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$ is an eigenvector of A if, for some scalar λ ,

$$A\mathbf{x} = \lambda\mathbf{x} \quad (86)$$

Where λ is the eigenvalue of A corresponding to the eigenvector $\mathbf{x}$.

Definition 12.3.1.2: The equation $\det(\lambda I - A) = 0$ is called the characteristic equation of A . In general, this is a polynomial where n is in the order of A .

$$p(\lambda) = \lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + \dots + c_n \quad (87)$$

Theorem 12.3.1.1: If A is an $n \times n$ matrix, then λ is an eigenvalue of A iff it satisfies $\det(\lambda I - A) = 0$.

Proof: By definition, λ is an eigenvalue of A iff $A\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq \mathbf{0}$. We have

$$A\mathbf{x} = \lambda\mathbf{x} \iff A\mathbf{x} = \lambda I\mathbf{x} \iff (\lambda I - A)\mathbf{x} = \mathbf{0} \quad (88)$$

By theorem about invertible matrices, the last equation has solution $\mathbf{x} \neq \mathbf{0}$ iff $\det(\lambda I - A) = 0$.

□

Definition 12.3.1.3: For a matrix A with eigenvalue λ , the null space of $\lambda I - A$ is called its eigenspace where the non-zero vectors in this space are the eigenvectors corresponding to λ .

Let λ_0 be an eigenvalue of a matrix A :

- **Definition 12.3.1.4:** The algebraic multiplicity of λ_0 is the power k with which $(\lambda - \lambda_0)$ appears as a factor of $\det(\lambda I - A)$.
- **Definition 12.3.1.5:** The geometric multiplicity of λ_0 is the dimension of the eigenspace corresponding to λ_0 .

Definition 12.3.1.6: Principle component analysis is the process of identifying the principle components (most important eigenvectors) of a data set. It takes a data set in n dimensions and outputs an $n \times n$ matrix, demonstrating how each dimension varies with each other dimension (i.e., it captures the variance of one dimension of the data with another):

1. Pretend we have some covariance matrix $COV_{n \times n}$.
2. Find the n eigenvalues and corresponding eigenvectors of $COV_{n \times n}$.
3. Order the eigenvalues in decreasing order to get the importance of each.
4. Work out what percentage of the total value each eigenvalue contributes and take the top k values, contributing to 80% of the total.
5. Let $P_{n \times k}$ be a matrix with the k chosen eigenvectors as column vectors.

12.3.2 Complex Matrices

For any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$ and a scalar $k \in \mathbb{C}$, the following holds:

- $\overline{\overline{\mathbf{u}}} = \mathbf{u}$.
- $\overline{k\mathbf{u}} = \overline{k}\overline{\mathbf{u}}$.
- $\overline{\mathbf{u} + \mathbf{v}} = \overline{\mathbf{u}} + \overline{\mathbf{v}}$.
- $\overline{\mathbf{u} - \mathbf{v}} = \overline{\mathbf{u}} - \overline{\mathbf{v}}$.

If A is an $m \times k$ complex matrix and B is a $k \times n$ matrix, then:

- $\overline{\overline{A}} = A$.
- $\overline{A^T} = (\overline{A})^T$.
- $\overline{AB} = \overline{A}\overline{B}$.

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ and a scalar $k \in \mathbb{C}$, the following holds:

- $\mathbf{u} \cdot \mathbf{v} = \overline{\mathbf{v}} \cdot \overline{\mathbf{u}}$.
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$.
- $(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$ and $\mathbf{u} \cdot (k\mathbf{v}) = \overline{k}(\mathbf{u} \cdot \mathbf{v})$.
- $\mathbf{v} \cdot \mathbf{v} \geq 0$ and $\mathbf{v} \cdot \mathbf{v} = 0$ iff $\mathbf{v} = \mathbf{0}$.

Theorem 12.3.2.1: If λ is an eigenvalue of a real $n \times n$ matrix A and $\mathbf{x}$ is a corresponding eigenvector, then $\overline{\lambda}$ is also an eigenvalue of A and $\overline{\mathbf{x}}$ is a corresponding eigenvector.

Proof: Since A is real, we have $A\overline{\mathbf{x}} = \overline{A\mathbf{x}} = \overline{\lambda\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}}$. □

Theorem 12.3.2.2: If A is a real symmetric matrix, then all (complex) eigenvalues of A are real.

Proof: Let $\lambda \in \mathbb{C}$ be an eigenvalue of A and $\mathbf{x} \in \mathbb{C}^n$ a corresponding eigenvector. Take the complex conjugate of both sides of the equation $A\mathbf{x} = \lambda\mathbf{x}$. We get $\overline{A\mathbf{x}} = \overline{\lambda\mathbf{x}}$, and, since $A = \overline{A}$, it follows that $A\overline{\mathbf{x}} = \overline{\lambda}\overline{\mathbf{x}}$. Then, using $A = A^T$, we compute the number $\overline{\mathbf{x}}^T A \mathbf{x}$ in two different ways:

$$\overline{\mathbf{x}}^T A \mathbf{x} = \overline{\mathbf{x}}^T (A\mathbf{x}) = \overline{\mathbf{x}}^T (\lambda\mathbf{x}) = \lambda(\overline{\mathbf{x}}^T \mathbf{x}) = \lambda(\mathbf{x} \cdot \mathbf{x}) = \lambda\|\mathbf{x}\|^2 \quad (89)$$

$$\overline{\mathbf{x}}^T A \mathbf{x} = (\overline{A\mathbf{x}})^T \mathbf{x} = (\overline{\lambda\mathbf{x}})^T \mathbf{x} = \overline{\lambda}(\overline{\mathbf{x}}^T \mathbf{x}) = \overline{\lambda}(\mathbf{x} \cdot \mathbf{x}) = \overline{\lambda}\|\mathbf{x}\|^2 \quad (90)$$

Since $\mathbf{x} \neq \mathbf{0}$, we have $\|\mathbf{x}\| \neq 0$, so $\lambda\|\mathbf{x}\|^2 = \overline{\lambda}\|\mathbf{x}\|^2$ implies $\lambda = \overline{\lambda}$, meaning $\lambda \in \mathbb{R}$. □

12.3.3 Inner Product Spaces

Definition 12.3.3.1: Let V be a real vector space. An inner product space on V is a function that associates to each pair $\mathbf{u}, \mathbf{v} \in V$ a real number $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{R}$, satisfying the following properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $k \in \mathbb{R}$:

- **Definition 12.3.3.2:** The symmetry axiom: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.
- **Definition 12.3.3.3:** The additivity axiom: $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$.
- **Definition 12.3.3.4:** The homogeneity axiom: $\langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$.
- **Definition 12.3.3.5:** The positivity axiom: $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ iff $\mathbf{v} = \mathbf{0}$.

Definition 12.3.3.6: Vectors $\mathbf{u}$ and $\mathbf{v}$ in an inner product space V are orthogonal if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Definition 12.3.3.7: Let W be a subspace in an inner product space V . Then the set

$$W^\perp = \{\mathbf{x} \in V \mid \langle \mathbf{x}, \mathbf{w} \rangle = 0 \forall \mathbf{w} \in W\} \quad (91)$$

is called the orthogonal complement of W .

Definition 12.3.3.8: Let $w_1, \dots, w_n \in \mathbb{R}^+$ be arbitrary numbers, which we'll call weights. The weighted Euclidean inner product on $\mathbb{R}^n$ is defined for vectors $\mathbf{u}, \mathbf{v}$ as:

$$\langle \mathbf{u}, \mathbf{v} \rangle = w_1 u_1 v_1 + w_2 u_2 v_2 + \dots + w_n u_n v_n \quad (92)$$

Definition 12.3.3.9: Let A be an invertible $n \times n$ matrix. Considering vectors in $\mathbb{R}^n$ as column vectors, define,

$$\langle \mathbf{u}, \mathbf{v} \rangle = A\mathbf{u} \cdot A\mathbf{v} \quad (93)$$

where the right-hand side uses the standard dot product in $\mathbb{R}^n$. This known as the inner product on $\mathbb{R}^n$ generated by A .

Definition 12.3.3.10: For vectors $\mathbf{p} = a_0 + a_1x + \dots + a_nx^n$ and $\mathbf{q} = b_0 + b_1x + \dots + b_nx^n$ in $\mathbb{P}_n$, define,

$$\langle \mathbf{p}, \mathbf{q} \rangle = a_0b_0 + a_1b_1 + \dots + a_nb_n \quad (94)$$

This is called the standard inner product on $\mathbb{P}_n$.

Definition 12.3.3.11: Fix distinct points $x_0, x_1, \dots, x_n \in \mathbb{R}$ (called sample points). For vectors $\mathbf{p} = p(x)$ and $\mathbf{q} = q(x)$ in $\mathbb{P}_n$, define

$$\langle \mathbf{p}, \mathbf{q} \rangle = p(x_0)q(x_0) + p(x_1)q(x_1) + \dots + p(x_n)q(x_n) \quad (95)$$

This is called the evaluation inner product at $x_0, x_1, \dots, x_n$.

Definition 12.3.3.12: Let $C[a, b]$ consist of all functions that are continuous on interval $[a, b]$. The operations in $C[a, b]$ are defined point-wise. For $\mathbf{f} = f(x)$ and $\mathbf{g} = g(x)$ in $C[a, b]$, define,

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_a^b f(x)g(x)dx \quad (96)$$

This is the inner product on the space $C[a, b]$.

Definition 12.3.3.13: For a complex vector $\mathbf{v}$, the Hermitian conjugation of $\mathbf{v}$, denoted $\mathbf{v}^\dagger$ is the conjugate transpose of the vector, i.e., $\mathbf{v}^\dagger = \overline{(\mathbf{v}^T)}$. A complex square matrix is a Hermitian matrix if it is equal to its own conjugate transpose.

Definition 12.3.3.14: Considering vectors in $\mathbb{C}^n$ as column vectors, the Hermitian inner product is defined as

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\dagger \mathbf{v} \quad (97)$$

Some standard (in)equalities for general inner products:

- **Definition 12.3.3.15:** Pythagoras' Theorem (inner-products), $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
- **Definition 12.3.3.16:** Cauchy-Schwarz Inequality (inner-products), $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|$.
- **Definition 12.3.3.17:** Triangle Inequality (inner-products), $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

12.3.4 The Gram-Schmidt Process

Definition 12.3.4.1: A set of vectors in an inner product space is called orthogonal if all pairs of distinct vectors in the set are orthogonal. An orthogonal set consisting of unit vectors is called orthonormal.

Theorem 12.3.4.1: If $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthogonal basis in inner product space V , then for any $\mathbf{u} \in V$,

$$\mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \dots + \frac{\langle \mathbf{u}, \mathbf{v}_n \rangle}{\|\mathbf{v}_n\|^2} \mathbf{v}_n \quad (98)$$

Moreover, if S is an orthonormal basis in V , then

$$\mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \dots + \langle \mathbf{u}, \mathbf{v}_n \rangle \mathbf{v}_n \quad (99)$$

Proof: If $\mathbf{u} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n$ then, since $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = 0$ for all $j \neq i$, we have that, for each i ,

$$\langle \mathbf{u}, \mathbf{v}_i \rangle = c_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle = c_i \|\mathbf{v}_i\|^2 \quad (100)$$

So $c_i = \frac{\langle \mathbf{u}, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2}$ as required.

□

Definition 12.3.4.2: The Projection Theorem states that if W is a subspace in a finite-dimensional inner product space V , then every vector $\mathbf{u} \in V$ can be uniquely expressed as $\mathbf{u} = \mathbf{w}_1 + \mathbf{w}_2$ where $\mathbf{w}_1 \in W$ and $\mathbf{w}_2 \in W^\perp$.

Definition 12.3.4.3: Every non-zero finite-dimensional inner product space V has an orthonormal basis. This can be found using the Gram-Schmidt process:

1. Let W be a subspace of V and let $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be a basis of W .
2. Let $\mathbf{v}_1 = \mathbf{u}_1$. Clearly $\{\mathbf{v}_1\}$ is an orthogonal basis for $W_1 = \text{span}(\mathbf{u}_1)$.
3. If $\{\mathbf{v}_1, \dots, \mathbf{v}_{r-1}\}$ is an orthogonal basis for $W_{r-1} = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_N)$, find a vector $\mathbf{v}_r$ such that $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is an orthogonal basis for W_r .

$$\mathbf{v}_r = \text{proj}_{W_{r-1}^\perp} \mathbf{u}_r = \mathbf{u}_r - \frac{\langle \mathbf{u}_r, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\langle \mathbf{u}_r, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 - \dots - \frac{\langle \mathbf{u}_r, \mathbf{v}_{r-1} \rangle}{\|\mathbf{v}_{r-1}\|^2} \mathbf{v}_{r-1} \quad (101)$$

4. Since $\mathbf{v}_r \in W_{r-1}^\perp$, the set $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is orthogonal.
5. Once all vectors have been found, they can be normalised to form an orthonormal basis for V .

Theorem 12.3.4.2: If V is a finite-dimensional inner product space, then any orthogonal set of vectors in V can be extended into either an orthogonal or orthonormal basis.

Proof: Let $a = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be an orthogonal set in V and $b = \{\mathbf{u}_{k+1}, \dots, \mathbf{u}_{k+n}\}$ some basis in V . Apply Gram-Schmidt to the b , and since a is an orthogonal set, we will have $\mathbf{v}_i = \mathbf{u}_i$ for $1 \leq i \leq k$. If $\mathbf{v}_r = \mathbf{0}$ at any step, do not add it to the output set. The final set will extend a , it will be orthogonal, and its span will be $\text{span}(a, b) = V$.

□

12.4 Decomposition

12.4.1 LU Decomposition

Definition 12.4.1.1: Matrix decomposition or factorisation is the process of representing a matrix as a product of two or more matrices with useful properties.

Definition 12.4.1.2: An elementary matrix is a matrix which differs from the identity matrix as a result of one elementary row operation (switching, addition, or multiplication).

Definition 12.4.1.3: LU decomposition is a method for solving linear systems by reducing the problem into two triangular systems. This method can be used on a square matrix A if U is its row echelon form, which was produced without any row exchanges being performed. The method is as follows:

1. Keep track of the row operations used to compute U by Gaussian elimination.
2. Let $E_1, \dots, E_k$ be the corresponding elementary matrices for each row operation.
3. Then the inverse matrices $(E_1)^{-1}, \dots, (E_k)^{-1}$ are easy to find.
4. Compute $L = (E_1)^{-1}(E_2)^{-1} \dots (E_k)^{-1}I$ by applying the corresponding row operations to the identity matrix.

Definition 12.4.1.4: A permutation matrix is a square matrix P obtained from I by permuting its rows.

Definition 12.4.1.5: A PLU decomposition of a square matrix A is a representation $A = PLU$ where P is a permutation matrix, L is lower triangular, and U is upper triangular.

12.4.2 Eigendecomposition

Definition 12.4.2.1: Eigendecomposition is the process of decomposing a matrix into PDP^{-1} form where P are eigenvectors and D is a diagonal matrix of eigenvalues.

Definition 12.4.2.2: Square matrices A and B are called similar if $A = P^{-1}BP$ for some invertible P . These matrices will have the same determinant, rank, trace, nullity, eigenvalues, etc.

Theorem 12.4.2.1: Matrices A and B are similar iff they represent the same linear operator $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, possibly in different bases.

Proof: Proof omitted.

□

Definition 12.4.2.3: A matrix A is called diagonalisable if it is similar to a diagonal matrix, i.e., there exists an invertible matrix P such that $P^{-1}AP$ is diagonal. Then P is said to diagonalise A .

Theorem 12.4.2.2: An $n \times n$ matrix is diagonalisable iff it has n linearly independent eigenvectors.

Proof: ($\implies$). Assume that there is an invertible matrix P and a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ such that $D = P^{-1}AP$, or $AP = PD$. Denote the i column vectors of P as $\mathbf{p}_i$, then

$$AP = A[\mathbf{p}_1 | \dots | \mathbf{p}_n] = [A\mathbf{p}_1 | \dots | A\mathbf{p}_n] \quad (102)$$

On the other hand,

$$PD = [\lambda_1 \mathbf{p}_1 | \dots | \lambda_n \mathbf{p}_n] \quad (103)$$

Since $AP = PD$, we conclude that $A\mathbf{p}_i = \lambda_i \mathbf{p}_i$ for all $1 \leq i \leq n$. Since P is invertible, its rank is n and so the vectors $\mathbf{p}_1, \dots, \mathbf{p}_n$ are linearly independent, making each of them an eigenvector. $\square$

($\impliedby$). Assume that A has n linearly independent eigenvectors $\mathbf{p}_1, \dots, \mathbf{p}_n$ and let $\lambda_1, \dots, \lambda_n$ be the corresponding eigenvalues. Define

$$P = [\mathbf{p}_1 | \dots | \mathbf{p}_n] \quad \text{and} \quad D = \text{diag}(\lambda_1, \dots, \lambda_n) \quad (104)$$

Then

$$AP = A[\mathbf{p}_1 | \dots | \mathbf{p}_n] = [A\mathbf{p}_1 | \dots | A\mathbf{p}_n] = [\lambda_1 \mathbf{p}_1 | \dots | \lambda_n \mathbf{p}_n] = PD \quad (105)$$

The columns of P are linearly independent, so its rank is n and it is invertible. Finally, $AP = PD$ is equivalent to $D = P^{-1}AP$. $\square$

To diagonalise a matrix A , we apply the following algorithm:

1. Find the eigenvalues of A .
2. Find a basis in each eigenspace of A and merge these bases into one set S .
3. If S has fewer than n vectors then A is not diagonalisable.
4. Else, form the matrix $P = [\mathbf{p}_1 | \dots | \mathbf{p}_n]$ where $S = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$. The set S contains n vectors and is linearly independent so is a basis for $\mathbb{R}^n$, hence P is invertible.
5. The matrix $D = P^{-1}AP$ is diagonal, $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ where, for each i , λ_i is the eigenvalue corresponding to $\mathbf{p}_i$.

Definition 12.4.2.4: Let A be a diagonalisable matrix. We have $A = PDP^{-1}$. Since the factors in this decomposition are made of eigenvectors and eigenvalues, it is called an eigendecomposition of A .

Theorem 12.4.2.3: If vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are eigenvectors of a matrix A corresponding to pairwise distinct eigenvalues $\lambda_1, \dots, \lambda_k$, then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly independent.

Proof: Let $r \leq k$ be the largest number such that $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent. Assume for contradiction that $r < k$, so $\{\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}\}$ is linearly dependent:

$$c_1 \mathbf{v}_1 + \dots + c_r \mathbf{v}_r + c_{r+1} \mathbf{v}_{r+1} = \mathbf{0} \quad (106)$$

where not all $c_1, \dots, c_r, c_{r+1}$ are 0. Since $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent, we conclude that $c_{r+1} \neq 0$. Since $\mathbf{v}_{r+1}$ is an eigenvector, we conclude that $c_i \neq 0$ for some $i \leq r$. We assumed that $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent, but $\{\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}\}$ is not. We derived that $c_{r+1} \neq 0$ and $c_i \neq 0$ for some $i \leq r$. Left multiply both sides of the linear combination by A and use $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$:

$$c_1 \lambda_1 \mathbf{v}_1 + \dots + c_r \lambda_r \mathbf{v}_r + c_{r+1} \lambda_{r+1} \mathbf{v}_{r+1} = \mathbf{0} \quad (107)$$

Instead multiplying the original equation by λ_{r+1} and subtracting the result, we get

$$c_1 (\lambda_1 - \lambda_{r+1}) \mathbf{v}_1 + \dots + c_r (\lambda_r - \lambda_{r+1}) \mathbf{v}_r = \mathbf{0} \quad (108)$$

So $c_i (\lambda_i - \lambda_{r+1}) = 0$, and hence $c_i = 0$, for all $i \leq r$, a contradiction. $\square$

12.4.3 QR Decomposition

Theorem 12.4.3.1: If A is an $m \times n$ matrix with linearly independent column vectors, then it can be factored as $A = QR$ where Q has orthonormal columns and R is an invertible, upper-triangular matrix.

Proof: Let A be an $m \times n$ matrix with linearly independent columns $\mathbf{u}_1, \dots, \mathbf{u}_n$. Let Q be the matrix formed from the column vectors given by the orthonormal set $\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ obtained by applying Gram-Schmidt to $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$. Or, in matrix form:

$$[\mathbf{u}_1 | \dots | \mathbf{u}_n] = [\mathbf{q}_1 | \dots | \mathbf{q}_n] \begin{pmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_1 \rangle \\ \langle \mathbf{u}_1, \mathbf{q}_2 \rangle & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{q}_n \rangle & \langle \mathbf{u}_2, \mathbf{q}_n \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_n \rangle \end{pmatrix} = QR \quad (109)$$

From Gram-Schmidt, for each $j \geq 2$, $\mathbf{q}_j$ is orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$. Hence,

$$R = \begin{pmatrix} \langle \mathbf{u}_1, \mathbf{q}_1 \rangle & \langle \mathbf{u}_2, \mathbf{q}_1 \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_1 \rangle \\ 0 & \langle \mathbf{u}_2, \mathbf{q}_2 \rangle & \dots & \langle \mathbf{u}_n, \mathbf{q}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \langle \mathbf{u}_n, \mathbf{q}_n \rangle \end{pmatrix} \quad (110)$$

□

Definition 12.4.3.1: The least squares problem states that, given a linear system $A\mathbf{x} = \mathbf{b}$ with m equations and n variables, find a vector $\mathbf{x}$ that minimises $\|\mathbf{b} - A\mathbf{x}\|$ (w.r.t. to the Euclidean inner product on $\mathbb{R}^n$). We call such a vector $\mathbf{x}$ a least squares solution to the system, the vector $\mathbf{b} - A\mathbf{x}$ is the least squares error vector, and the number $\|\mathbf{b} - A\mathbf{x}\|$ is the least squares error.

Theorem 12.4.3.2: If W is a finite-dimensional subspace in an inner product space V and $\mathbf{b} \in V$, then $\text{proj}_W \mathbf{b}$ is the best approximation to $\mathbf{b}$ from W in the sense that

$$\|\mathbf{b} - \text{proj}_W \mathbf{b}\| \leq \|\mathbf{b} - \mathbf{w}\| \quad (111)$$

For each vector $\mathbf{w} \in W$, and the inequality is strict for all $\mathbf{w} \neq \text{proj}_W \mathbf{b}$.

Proof: For any vector $\mathbf{w} \in W$, write

$$\mathbf{b} - \mathbf{w} = (\mathbf{b} - \text{proj}_W \mathbf{b}) + (\text{proj}_W \mathbf{b} - \mathbf{w}) \quad (112)$$

By Pythagoras' theorem, we have,

$$\|\mathbf{b} - \mathbf{w}\|^2 = \|\mathbf{b} - \text{proj}_W \mathbf{b}\|^2 + \|\text{proj}_W \mathbf{b} - \mathbf{w}\|^2 \geq \|\mathbf{b} - \text{proj}_W \mathbf{b}\|^2 \quad (113)$$

Moreover, the inequality is strict whenever $\mathbf{w} \neq \text{proj}_W \mathbf{b}$.

□

We can find a least squares solution by:

1. Let $W = \{A\mathbf{x} | \mathbf{x} \in \mathbb{R}^n\}$ be the column space for A .
2. Since $\text{proj}_W \mathbf{b}$ is the best approximation to $\mathbf{b}$ from W , the least squares solutions to $A\mathbf{x} = \mathbf{b}$ are exactly solutions to

$$A\mathbf{x} = \text{proj}_W \mathbf{b} \quad (114)$$

3. The representation $\mathbf{b} = \mathbf{w}_1 + \mathbf{w}_2$, $\mathbf{w}_1 \in W$, $\mathbf{w}_2 \in W^\perp$ is unique, so (114) is equivalent to the condition $\mathbf{b} - A\mathbf{x} \in W^\perp$.

4. The columns of A are the rows of A^T , so the condition can be rewritten as,

$$A^T A\mathbf{x} = A^T \mathbf{b} \quad (115)$$

Definition 12.4.3.2: The above equation is the normal equation associated with $A\mathbf{x} = \mathbf{b}$.

12.4.4 Linear Regression

Theorem 12.4.4.1: For any $m \times n$ matrix A , A has linearly independent columns iff $A^T A$ is invertible.

Proof: Both the columns of A being linearly independent and $A^T A$ being square are dependent on $A\mathbf{x} = \mathbf{0}$ only having one trivial solution. Each solution of $A\mathbf{x} = \mathbf{0}$ is a solution of $A^T A\mathbf{x} = \mathbf{0}$. Let $\mathbf{x}_0$ be a solution, then $A\mathbf{x}_0$ is both in the column space of A and in the null space of A^T , hence $A\mathbf{x}_0 = \mathbf{0}$. Thus $A\mathbf{x} = \mathbf{0}$ only has a trivial solution iff the same is true for $A^T A\mathbf{x} = \mathbf{0}$. □

To find a unique least squares solution, we let A be an $m \times n$ invertible matrix with linearly independent column vectors.

1. From the previous theorem, for every column $\mathbf{b} \in \mathbb{R}^m$, the least squares solution to $A\mathbf{x} = \mathbf{b}$ is unique:

$$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b} \quad (116)$$

2. For the last item, substitute $A = QR$ into the first equation above and simplify using $Q^T Q = I$ as Q has orthonormal columns.
3. If $A = QR$ is a QR-decomposition, then,

$$\mathbf{x} = R^{-1} Q^T \mathbf{b} \quad (117)$$

12.4.5 Spectral Decomposition

Definition 12.4.5.1: A square matrix Q is known as an orthogonal matrix if $Q^T = Q^{-1}$.

Theorem 12.4.5.1: For any $n \times n$ matrix Q , the following are equivalent:

1. Q is orthogonal.
2. The rows of Q form an orthogonal basis in $\mathbb{R}^n$.
3. The columns of Q form an orthogonal basis in $\mathbb{R}^n$.

Proof: For (1) $\iff$ (3): If $Q = [\mathbf{u}_1 | \dots | \mathbf{u}_n]$, then each entry (i, j) in the product $Q^T Q$ is $\langle \mathbf{u}_i, \mathbf{u}_j \rangle$. Hence, $Q^T Q = I$ iff we have $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for all $i \neq j$ and $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$ for all i , which holds iff the set $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is orthonormal. Any orthonormal set with n vectors in $\mathbb{R}^n$ is a basis. □

Properties of an orthogonal matrix:

- The transpose of an orthogonal matrix is also orthogonal.
- The inverse of an orthogonal matrix is also orthogonal.
- A product of orthogonal matrices is also orthogonal.
- If Q is orthogonal then $\det(Q) = \pm 1$.

Definition 12.4.5.2: Let A and B be two similar matrices such that $P^{-1}AP = B$ where B is also a diagonal matrix. If P can be chosen to be orthogonal, then A and B are orthogonally similar, and A is orthogonally diagonalisable.

Theorem 12.4.5.2: (The Spectral Theorem) For any $n \times n$ matrix A , the following are equivalent:

1. A is orthogonally diagonalisable.
2. A has an orthonormal set of n eigenvectors.
3. A is symmetric.

Proof: (1) $\iff$ (2) is the same as for general diagonalisation.

(1) $\implies$ (3), If $A = QDQ^T$, then $A^T = (QDQ^T)^T = (Q^T)^T D^T Q^T = QDQ^T = A$.

□

Theorem 12.4.5.3: If A is a symmetric matrix, then:

1. All eigenvalues of A are real.
2. Eigenvectors from different eigenspaces are orthogonal.

Proof: Proof for (1) is omitted.

(2) Assume that λ_1 and λ_2 are two different eigenvalues of A with corresponding eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$. We have:

$$\lambda_1 \langle \mathbf{u}_1, \mathbf{u}_2 \rangle = \langle A\mathbf{u}_1, \mathbf{u}_2 \rangle \quad (118)$$

Expanding the dot product, we get that

$$(\mathbf{u}_1)^T A^T \mathbf{u}_2 = \langle \mathbf{u}_1, A\mathbf{u}_2 \rangle = \lambda_2 \langle \mathbf{u}_1, \mathbf{u}_2 \rangle \quad (119)$$

Since $(\lambda_1 - \lambda_2) \langle \mathbf{u}_1, \mathbf{u}_2 \rangle = 0$ and $\lambda_1 \neq \lambda_2$, we get $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = 0$ as required.

□

Orthogonally diagonalising a symmetric matrix:

1. Let A be a symmetric matrix.
2. Find the eigenvalues and a basis in each eigenspace of A .
3. Apply Gram-Schmidt to each basis to obtain an orthonormal basis for each eigenspace.
4. Form the matrix Q , whose columns are the vectors from Step 3. Q will orthogonally diagonalise A , and the diagonal $D = Q^T A Q$ contains the eigenvalues of A in the same order as the eigenvectors in Q .

Definition 12.4.5.3: Spectral decomposition: Let A be a symmetric matrix, $\mathbf{u}_1, \dots, \mathbf{u}_n$ be an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A , with $\lambda_1, \dots, \lambda_n$ the corresponding eigenvalues.

$$A = \lambda_1 \mathbf{u}_1 (\mathbf{u}_1)^T + \lambda_2 \mathbf{u}_2 (\mathbf{u}_2)^T + \dots + \lambda_n \mathbf{u}_n (\mathbf{u}_n)^T \quad (120)$$

Part V

Discrete Mathematics (Elective)

13 Introduction to Discrete Mathematics

13.1 Arrangements and Permutations

13.1.1 Arrangements

Definition 13.1.1.1: An arrangement is a sequence of items in which the order of the items in the sequence matters.

Definition 13.1.1.2: The length of an arrangement is the number of items in the sequence.

Standard Problem 1: How many arrangements of length k can be made from elements of a set S of size $|S| = n$?

Solution: n^k as arrangement has k places, each of which can be occupied by any one of n entries.

Example 13.1.1.1: How many four-letter “words” can be made from the standard A-Z alphabet?

Solution: $26^4 = 456,796$ words.

Standard Problem 2: How many arrangements of length k with no repeats can be made from elements of a set S of size $|S| = n$?

Solution:

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!}, \quad k \leq n \quad (121)$$

As we look at the available choices for each successive item in the list, the number of options is reduced by one each time.

Definition 13.1.1.3: The multiplication principle states that, for an arrangement of length k , if for $i \in \{1, 2, \dots, k\}$ there are n_i choices for the n^{th} item, regardless of previous choices, then the number of possible arrangements is $n_1 n_2 \dots n_k$.

13.1.2 Permutations

Definition 13.1.2.1: A permutation on a finite set S is an arrangement of length $|S|$ that uses each item from S exactly once.

Proposition 13.1.2.1: If $|S| = n$, then there are $n!$ permutations on S .

Proof: This is Standard Problem 2 with $n = k$.

□

Definition 13.1.2.2: We generalise this concept as, for a non-negative integer r , an r -permutation on S is an arrangement of r distinct elements from S . We write $P(n, r)$ for the number of r -permutations on a set of size n , which, by Standard Problem 2, we have,

$$P(n, r) = n(n-1)\dots(n-r+1) = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n \quad (122)$$

13.1.3 Counting Examples

Definition 13.1.3.1: The addition principle states that, for two disjoint, finite sets A and B . Then,

$$A \cap B = \emptyset \quad |A \cup B| = |A| + |B| \quad (123)$$

More generally, if $A_1, A_2, \dots, A_n$ are pairwise disjoint, then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n| \quad (124)$$

13.2 Combinations

Definition 13.2.0.1: A combination from a set of objects is an unordered selection of objects of that set.

Standard Problem 3: How many combinations without repetition do we have of a given size k from a set of size n ?

Solution: First consider that there are $P(n, k) = \frac{n!}{(n-k)!}$ arrangements without repetition. These are ordered lists, sorting them into groups of lists using the same letters will give groups of size $k!$. This means that the number of combinations is

$$\frac{P(n, k)}{k!} = \frac{n!}{k!(n-k)!} \quad (125)$$

Using the binomial coefficient notation, this means the number of combinations is $\binom{n}{k}$.

Example 13.2.0.1: How many ways are there of selecting ten different letters of the alphabet?

Solution: $\binom{26}{10} = \frac{26!}{10!16!}$.

Proposition 13.2.0.1: $\binom{n}{k} = \binom{n}{n-k}$

Proof: This can be seen from the formula for the binomial. □

Proposition 13.2.0.2: Pascal's formula, for $1 \leq k \leq (n-1)$, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Proof: Proof excluded. □

Theorem 13.2.0.1: The binomial theorem, if n is a positive integer and $a, b \in \mathbb{R}$, then

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad (126)$$

Proof: The expression $(a+b)(a+b)\dots(a+b)$ when multiplied out, gives various terms of the form $a^k b^{n-k}$. For fixed k , the number of terms $a^k b^{n-k}$ is $\binom{n}{k}$ since one chooses the k brackets in the product form from which to choose the factor a rather than b . □

13.3 Arrangements and Combinations with Repetitions

13.3.1 Arrangements with Repetitions

Standard Problem 4: How many arrangements are there of a list of objects in which some objects are repeated?

Solution: **Theorem 13.3.1.1:** Given a list of n objects of r different types, in which objects of type i appear n_i times, the number of arrangements is

$$P(n; n_1, n_2, \dots, n_r) = \frac{n!}{n_1! n_2! \dots n_r!} \quad (127)$$

Proof: Proof omitted. □

Example 13.3.1.1: How many arrangements of REARRANGE are there, and of those, how many of the arrangements have at least two R's together?

Solution: The letters to arrange are RRR AA EE G and N. Applying Theorem 13.3.1.1 with $n = 9$ and $n_1 = 3, n_2 = 2, n_3 = 2, n_4 = 1, n_5 = 1$, we get

$$\frac{9!}{3!2!2!} = 15,120 \quad (128)$$

We can then take all the cases where all the R's are separate. To do this, we arrange all the other letters, giving $\frac{6!}{2!2!}$ arrangements. We then insert the R's into the available gaps, giving 7 possible spots for 3 R's, meaning the number of arrangements are

$$\frac{9!}{3!2!2!} - \binom{7}{3} \left(\frac{6!}{2!2!} \right) = 8,820 \quad (129)$$

13.3.2 Combinations with Repetitions

Standard Problem 5: Given a set of n objects, how many combinations of size k can be chosen if objects may be selected more than once? (How many k -combinations with repetitions from n objects are there?)

Solution: To solve this problem, first consider

Standard Problem 6: How many k -combinations with repetitions from n objects are there in which each object is chosen at least once in the combination?

Solution: **Theorem 13.3.2.1:** The number of k -combinations from n different objects with repetition and each object occurring at least once is $\binom{k-1}{n-1}$ for $k \geq n$.

Proof: This question clearly has a non-zero answer only if $k \geq n$. Since order does not matter in our combinations, we represent each combination with its items in non-decreasing order. Now insert markers into the strings at the points where the numbers change. Because we have specified that each number appears at least once, there must be $(n-1)$ markers occupying $(k-1)$ possible positions in each of these lists.

□

Example 13.3.2.1: Count the 5-combinations with repetitions from $\{1, 2, 3\}$ in which each number appears at least once.

Solution: Using the method above, we get the following combinations:

$$\begin{array}{ll} 11123 & \leftrightarrow 111|2|3 \\ 11223 & \leftrightarrow 11|22|3 \\ 11233 & \leftrightarrow 11|2|33 \\ 12223 & \leftrightarrow 1|222|3 \\ 12233 & \leftrightarrow 1|22|33 \\ 12333 & \leftrightarrow 1|2|333 \end{array}$$

There are 4 possible gaps for 2 markers, so the number of ways of arranging the markers is $\binom{4}{2} = 6$.

We can adapt this solution to solve Standard Problem 5 by allowing the markers to also be placed next to one another, or outside of the item strings. Taking this into account, we get:

Theorem 13.3.2.2: The number of k -combinations from n different objects with repetition is

$$\binom{k+n-1}{n-1} = \binom{k+n-1}{k} \quad (130)$$

Proof: As explained above.

□

13.3.3 The Extended Binomial Coefficients

Proposition 13.3.3.1: The principle of upper negation states that, if n and k are positive integers, then

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k} = (-1)^k \binom{k+n-1}{n-1} \quad (131)$$

Proof:

$$\binom{-n}{k} = \frac{(-n)(-n-1)(-n-2)\dots(-n-k+1)}{k!} \quad (132)$$

Factoring out a factor of -1,

$$\binom{-n}{k} = (-1)^k \frac{(n)(n+1)(n+2)\dots(n+k-1)}{k!} \quad (133)$$

Which simplifies to give the claimed result.

□

14 Combinatorics

14.1 The Three Principles

14.1.1 Mathematical Induction

Definition 14.1.1.1: Mathematical induction provides a method for establishing the truth of a sequence of statements, each of which is associated with a non-negative integer. In its simplest form, it gives efficient proofs of identities such as the binomial formula.

Theorem 14.1.1.1: The principle of mathematical induction states that: suppose that $P(n)$ is a sequence of propositions labelled by integers $n \geq n_0$. Suppose that

- a. $P(n_0)$ is true, and
- b. For any $n \geq n_0$, the truth of $P(n)$ implies the truth of $P(n+1)$.

Then, for every $n \geq n_0$, $P(n)$ is true.

Proof: $P(n_0)$ is true by assumption (a). Then by (b), $P(n_0+1)$ is also true. Again, by (b), $P(n_0+2)$ is also true, and so on. This chain of deductions eventually reaches any given $n \geq n_0$, so $P(n)$ is true for all $n \geq n_0$.

□

14.1.2 The Pidgeon-Hole Principle

Definition 14.1.2.1: The pidgeon-hole principle states that if n envelopes are distributed among m pidgeon-holes, and $n \geq m$, then at least one pidgeon-hole contains more than one envelope.

14.1.3 The Inclusion-Exclusion Principle

Proposition 14.1.3.1: We have

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3| \quad (134)$$

Proof: Apply $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$ repeatedly.

□

Theorem 14.1.3.1: The inclusion-exclusion formula states that, for a positive integer n and finite sets $A_1, \dots, A_n$,

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{k=1}^n \left((-1)^{k+1} \times \sum_{i_1 < i_2 < \dots < i_k} |A_{i_1} \cup A_{i_2} \cup \dots \cup A_{i_k}| \right) \quad (135)$$

Proof: A slightly tricky induction on n , with the base case $n = 2$ given by $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. □

Definition 14.1.3.1: A derangement of the numbers $\{1, 2, \dots, n\}$ is a permutation in which no number is in its correct position.

Proposition 14.1.3.2: The number of derangements of $1, 2, \dots, n$ is given by

$$d(n) = \sum_{k=0}^n (-1)^k \frac{n!}{k!} \quad (136)$$

Proof: We subtract from $n!$ the number of non-derangements to get

$$d(n) = n! \sum_{k=0}^n (-1)^k \frac{1}{k!} \quad (137)$$

□

In particular, the proportion of permutations that are derangements is

$$\frac{d(n)}{n!} = \sum_{k=0}^n (-1)^k \frac{1}{k!} \quad (138)$$

This is the power series expansion of e^{-1} , so

$$\lim_{n \rightarrow \infty} \frac{d(n)}{n!} = \frac{1}{e} \quad (139)$$

14.2 Recurrence Relations

14.2.1 Basic Definitions

Definition 14.2.1.1: As defined by (Perepelitsa, 2020, p.1), a recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence for all integers n with $n \geq n_0$, where n_0 is a non-negative integer.

Definition 14.2.1.2: As defined by (Perepelitsa, 2020, p.1), a sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

Definition 14.2.1.3: As defined by (Perepelitsa, 2020, p.1), the initial conditions for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Let $\{a_n\}$ be a sequence indexed by integers $n = 0, 1, 2, \dots$, we are given:

- The initial conditions $a_0, a_1, \dots, a_{r-1}$.
- The recurrence relation

$$a_n = \sum_{i=1}^r c_i a_{n-i} = c_1 a_{n-1} + \dots + c_r a_{n-r}, \quad (n \geq r) \quad (140)$$

where $c_1, \dots, c_r$ are specified constants.

Definition 14.2.1.4: The recurrence relation is linear because no term involves more than a single factor of the a_i , and it is homogeneous because every term involves an a_i .

Definition 14.2.1.5: We seek solutions to the recurrence relation (140) of the form $a_n = x^n$, $x \neq 0$. Substituting this into the relation, we get

$$x^n = \sum_{i=1}^r c_i x^{n-i}, \quad (n \geq r) \quad (141)$$

Dividing through by x^{n-r} , we get

$$x^r - \sum_{i=1}^r c_i x^{r-i} = 0 \quad (142)$$

This is the characteristic equation.

14.2.2 General Method for Solving Homogeneous Linear Recurrence Relations

1. Write down the characteristic equation of the recurrence relation as in (142).
2. Find the equation's roots $\alpha_1, \alpha_2, \dots, \alpha_r$.
3. Then, for each i , $a_n = d(\alpha_i)^n$ satisfies the recurrence relation (140) for any constant d .
4. By linearity, any combination of the $(\alpha_i)^n$ also solves the recurrence relation.
5. Find the general solution to the recurrence relation.

- If the roots are distinct, then the general solution to the recurrence relation is:

$$a_n = g(n) = d_1(\alpha_1)^n + d_2(\alpha_2)^n + \dots + d_r(\alpha_r)^n \quad (143)$$

for arbitrary constants $d_1, d_2, \dots, d_r$.

- If α is a root repeated k times, then in place of the expression (143), we use instead

$$a_n = g(n) = (e_1 + e_2 n + \dots + e_k n^{k-1})(\alpha_1)^n + d_2(\alpha_2)^n + \dots + d_r(\alpha_r)^n \quad (144)$$

for constants $e_1, e_2, \dots, e_k$.

6. Apply the initial conditions, giving r linear equations in the r unknown constants d_i . Solve these to get a solution a_n .

Example 14.2.2.1: Solve $a_n = 3a_{n-1} + 4a_{n-2}$, $(n \geq 2)$ where $a_0 = 1$ and $a_1 = 0$.

Solution: The characteristic equation is

$$x^2 - 3x - 4 = 0 \quad \rightarrow \quad (x - 4)(x + 1) = 0 \quad (145)$$

This has distinct roots $x = 4$ and $x = -1$, so the general solution is

$$a_n = d_1(-1)^n + d_2(4)^n \quad (146)$$

Using the initial conditions gives

$$1 = a_0 = d_1 + d_2 \quad \text{and} \quad 0 = a_1 = -d_1 + 4d_2$$

Hence $d_1 = \frac{4}{5}$ and $d_2 = \frac{1}{5}$, so the solution is

$$a_n = \frac{1}{5}(4^n) + \frac{4}{5}(-1)^n \quad (147)$$

14.2.3 General Method for Solving Inhomogeneous Linear Recurrence Relations

Redefine the recurrence relation (140) to include an inhomogeneous part $f(n)$,

$$a_n = c_1 a_{n-1} + \dots + c_r a_{n-r} + f(n) \quad (148)$$

$f(n)$	Form of Particular Solution $p(n)$
A	B_0 (constant)
An	$B_0 + B_1 n$ (linear)
An^2	$B_0 + B_1 n + B_2 n^2$ (quadratic)
Aq^n	$B_0 q^n$ (exponential)
Anq^n	$(B_0 + B_1 n)q^n$ (etc...)

Table 1: Forms of the Particular Solution

- Find the general solution to the homogeneous problem $g(n) = a_n$.
- Find a specific particular solution $p(n) = a_n$ to the full inhomogeneous problem using Table 1.
 - If $f(n)$ contains q^n where q is a root of the characteristic equation, then q^n should be replaced by nq^{n-1} .
 - If 1 is a root of the characteristic equation, then $f(n) = An^k$ should be considered as $f(n) = An^k \cdot 1^n$ and so should be treated as above.
- The general solution to the inhomogeneous problem is then the sum of these

$$a_n = g(n) + p(n) \quad (149)$$

- Apply the initial conditions to determine the unknown constants.

Example 14.2.3.1: Solve $a_n = a_{n-1} + 2a_{n-2} - 2n$, ($n \geq 2$) where $a_0 = a_1 = 0$.

Solution: Solving the homogeneous recurrence relation, we get the characteristic equation

$$x^2 - x - 2 = 0 \quad \rightarrow \quad (x - 2)(x + 1) = 0 \quad (150)$$

Which has distinct roots $x = -1$ and $x = 2$, giving the general solution

$$g(n) = d_1(-1)^n + d_2(2)^n \quad (151)$$

The inhomogeneous part is $f(n) = -2n$, so we seek a particular solution of the form

$$p(n) = B_0 + B_1 n \quad (152)$$

Setting $a_n = p(n)$ in the inhomogeneous recurrence relation gives

$$B_1 n + B_0 = (B_1(n - 1) + B_0) + 2(B_1(n - 2) + B_0) - 2n \quad (153)$$

Which we solve to get $B_1 = 1$ and $B_0 = \frac{5}{2}$, meaning the particular solution is

$$p(n) = n + \frac{5}{2} \quad (154)$$

This gives us the general solution to the inhomogeneous problem to be

$$a_n = g(n) + p(n) = d_1(-1)^n + d_2(2)^n + n + \frac{5}{2} \quad (155)$$

Substituting in the initial conditions, we get that

$$a_n = -\frac{1}{2}(-1)^n - 2^n + n + \frac{5}{2} \quad (156)$$

14.3 Generating Functions

Definition 14.3.0.1: Generating functions encode combinatorial information into an analytical form so we can solve counting problems. For example, the ordinary generating function for the sequence $a_0, a_1, \dots$ is

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \quad (157)$$

14.3.1 Counting with Generating Functions

Example 14.3.1.1: How many solutions are there to the equation

$$e_1 + e_2 + \dots + e_n = k \quad (158)$$

where each $e_i \in \{0, 1\}$ and n, k are given, positive integers?

Solution: When constructing our generating function, each e_i is associated with the expression $(x^0 + x^1) = (1 + x)$, the generating function is then the n -fold product,

$$f(x) = (1 + x) \cdot (1 + x) \dots (1 + x) = (1 + x)^n \quad (159)$$

If we expand the generating function as a power series, we get

$$(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k \quad (160)$$

Each term x^k corresponds to a choice of x^{e_1} from the first bracket, x^{e_2} from the second bracket, etc., in which the total power of x is $e_1 + \dots + e_n = k$, i.e., a solution to (158).

Lemma 14.3.1.1: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $1, 1, \dots, 1, 0, 0, \dots$ (first $(n + 1)$ terms are 1) is

$$\sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x} \quad (161)$$

Proof: This is the sum of a geometric series which can be proven directly in the form

$$(1 - x) \sum_{k=0}^n x^k = \sum_{k=0}^n (x^k - x^{k+1}) = 1 - x^{n+1} \quad (162)$$

since the sum telescopes.

□

Lemma 14.3.1.2: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $1, 1, 1, \dots$ is

$$\sum_{k=0}^n x^k = \frac{1}{1 - x} \quad (163)$$

Proof: The sum to infinity makes analytical sense only for $|x| < 1$, in which case we can take $n \rightarrow \infty$ in (161) to get (163).

□

Lemma 14.3.1.3: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $\binom{n+k-1}{k}$, ($k \geq 0$) is

$$\sum_{k=0}^n \binom{n+k-1}{k} x^k = \frac{1}{(1-x)^n} \quad (164)$$

Proof: Recall from the extended form of the binomial theorem that

$$(1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k \quad (165)$$

As claimed. □

Lemma 14.3.1.4: For $n \geq 0$, $n \in \mathbb{Z}$, if $f(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g(x) = \sum_{k=0}^{\infty} b_k x^k$ are the generating functions of sequences a_k and b_k respectively, then

$$f(x)g(x) = \sum_{k=0}^{\infty} c_k x^k \quad (166)$$

is the generating function of the sequence $c_k = \sum_{l=0}^k a_l b_{k-l}$.

Proof: This is just collecting terms. □

Example 14.3.1.2: Find the coefficient of x^{17} in the expression $f(x) = (1 + x + x^2 + \dots + x^{10})^5$.

Solution: We use Lemma 14.3.1.2 and Lemma 14.3.1.4 to write

$$f(x) = \left(\frac{1-x^{11}}{1-x} \right)^5 = (1-x)^{-5} (1-x^{11})^5 \quad (167)$$

Considering separately the two terms in the product, we have

$$(1-x^{11})^5 = \sum_{i=0}^5 \binom{5}{i} (-1)^i x^{11i} \quad (168)$$

And

$$(1-x)^{-5} = \sum_{j=0}^{\infty} \binom{4+j}{4} x^j \quad (169)$$

The terms in the product

$$\left(\sum_{i=0}^5 \binom{5}{i} (-1)^i x^{11i} \right) \left(\sum_{j=0}^{\infty} \binom{4+j}{4} x^j \right) \quad (170)$$

that contribute x^{17} are $(i, j) = (0, 17), (1, 6)$, so the required coefficient is

$$\binom{5}{0} \binom{21}{4} - \binom{5}{1} \binom{10}{4} = 5,985 - 1,050 = 4,935 \quad (171)$$

14.3.2 Recurrence Relations and Generating Functions

Example 14.3.2.1: Solve for a_n satisfying the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$, ($n \geq 2$) with initial conditions $a_0 = 1$ and $a_1 = -1$.

Idea: Write the generating function for a_n as $f(x) = \sum_{n=0}^{\infty} a_n x^n$. In this sum, substitute in for a_n using the recurrence relation for those values of n for which it is valid, then tidy up the remaining terms using the initial conditions.

Solution: In this case, the recurrence relation allows us to replace a_n by $a_{n-1} + 2a_{n-2}$ for all $n \geq 2$, so we can write

$$f(x) = 1 - x + \sum_{n=2}^{\infty} a_{n-1} x^n + 2 \sum_{n=2}^{\infty} a_{n-2} x^n \quad (172)$$

We want to relate the two sums to $f(x)$ so we take out factors of x so that the index a_k matches the power x^k each time

$$f(x) = 1 - x + x \sum_{n=2}^{\infty} a_{n-1} x^{n-1} + 2x^2 \sum_{n=2}^{\infty} a_{n-2} x^{n-2} \quad (173)$$

We can then re-label the summation indices to get a_n everywhere

$$f(x) = 1 - x + x \sum_{n=1}^{\infty} a_n x^n + 2x^2 \sum_{n=0}^{\infty} a_n x^n \quad (174)$$

We can see that the last term is $\sum_{n=0}^{\infty} a_n x^n = f(x)$ by definition and the first summation is missing the first term, so we can rewrite the equation as

$$f(x) = 1 - x + x(f(x) - 1) + 2x^2 f(x) \quad (175)$$

Rearranging for $f(x)$, we get that

$$f(x) = \frac{1}{1+x} = 1 - x + x^2 - x^3 \dots \quad (176)$$

So our solution is $a_n = 1$ if n is even, and $a_n = -1$ if n is odd.

14.3.3 Partitions

Definition 14.3.3.1: A partition of a positive integer n is an expression of n as an unordered sum of positive integers, called the parts of the partition. We denote the number of partitions of n as $p(n)$.

Theorem 14.3.3.1: The number of partitions $p(n)$ of n is the coefficient of x^n in the power series expansion of the infinite product

$$\prod_{k=1}^{\infty} (1 - x^k)^{-1} = \sum_{n=0}^{\infty} x^n p(n) \quad (177)$$

Proof: Proof omitted.

□

14.3.4 Calculus of Generating Functions

Theorem 14.3.4.1: If a_n is a sequence with generating function $f(x) = \sum_{n=0}^{\infty} a_n x^n$, then the generating function for the sequence $b_n = na_n$ is $xf'(x)$.

Proof:

$$\sum_{n=0}^{\infty} na_n x^n = x \sum_{n=0}^{\infty} a_n n x^{n-1} = x \frac{d}{dx} \left(\sum_{n=0}^{\infty} a_n x^n \right) = x f'(x)$$

□

Theorem 14.3.4.2: If $f(x)$ is the generating function of the sequence a_n , then $\frac{f(x)}{1-x}$ is the generating function of the sequence $s_n = \sum_{k=0}^n a_k$.

Proof: Proof omitted.

□

15 Graph Theory

15.1 What is a Graph?

15.1.1 Basic Definitions

Definition 15.1.1.1: A simple graph G is a pair of sets $G = (V, E)$ where each element of E is a 2-element subset of V .

Definition 15.1.1.2: Given a graph $G = (V, E)$, if $|V|$ and $|E|$ are finite, then G is a finite graph.

Definition 15.1.1.3: Two vertices are adjacent if there is an edge between them, that is, $u, v \in V$ are adjacent if $\{u, v\} \in E$.

Definition 15.1.1.4: An edge is incident to a vertex if that vertex is one of the endpoints of the edge.

15.1.2 Simple Graphs

Definition 15.1.2.1: The empty/null graph on m vertices is called N_m where $|V| = m$ but $|E| = 0$.

Definition 15.1.2.2: The complete graph on n is called K_n . Given $|V| = n$, we include an edge between every pair of vertices, so $|E| = \binom{n}{2}$.

Definition 15.1.2.3: The cycle graph on n vertices is called C_n . Given $|V| = n$, the cycle graph is constructed by listing the vertices in some order $v_1, v_2, \dots, v_n$ and then adding the cycle of n edges $v_1v_2, \dots, v_{n-1}v_n, v_nv_1$.

Definition 15.1.2.4: The star graph on n vertices is called S_n . Of the $|V| = n$ vertices, $(n - 1)$ are leaves with a single edge to a central hub vertex.

Definition 15.1.2.5: The wheel graph on n vertices is called W_n . It can be constructed from the star graph S_n by adding a cycle around the leaves of S_n .

Definition 15.1.2.6: The n -cube has 2^n vertices and is denoted Q_n .

15.1.3 Handshaking

Lemma 15.1.3.1: For any graph $G = (V, E)$, we have $\sum_{v \in V} \deg(v) = 2|E|$.

Proof: Each edge contributes 2 to the sum of degrees, one to each vertex it is incident to.

□

Theorem 15.1.3.1: In any graph, the number of odd-degree vertices is even.

Proof: Let $|V| = n$ and $|E| = m$, and let the number of vertices of odd degree be k . By Lemma 15.1.3.1, we have that the sum of degrees is even, and furthermore the sum of the degrees of the $(n - k)$ vertices of even degree is a sum of even numbers, and thus is even.

Therefore, the sum of the degrees of the remaining k vertices of odd degree must also be even. As a sum of odd numbers, there must be an even number of terms for the sum to be even, and hence there must be an even number of vertices of odd degree.

□

15.1.4 Graph Isomorphism, Complements, and Subgraphs

Definition 15.1.4.1: Two simple graphs G_1 and G_2 are isomorphic, $G_1 \cong G_2$, if there exists a bijective function mapping the vertices from G_1 to the vertices of G_2 which preserves adjacency.

Definition 15.1.4.2: Let $G = (V, E)$ be a simple graph. Then the complement of G is $\overline{G} = (V, F)$ where $F = \{uv : u, v \in V \text{ and } uv \notin E\}$.

Definition 15.1.4.3: Given a graph $G = (V, E)$ and a graph $G_1 = (V_1, E_1)$ such that $V_1 \subseteq V$ and $E_1 \subseteq E$, then G_1 is a subgraph of G .

15.2 Eulerian Graphs

15.2.1 Travelling in Graphs

Definition 15.2.1.1: Given a graph $G = (V, E)$, a walk in G is a sequence of vertices $v_1, v_2, \dots, v_k$ from V together with edges $e_1, e_2, \dots, e_{k-1}$ in E such that $e_i = \{v_i, v_{i+1}\}$ for all $1 \leq i \leq (k-1)$ where the length of the walk is the number of edges. If $v_1 = v_k$, the walk is closed.

Definition 15.2.1.2: A trail is a sequence of vertices $v_1, v_2, \dots, v_k$ from V together with distinct edges $e_1, e_2, \dots, e_{k-1}$ in E such that $e_i = v_i v_{i+1}$. That is, a trail is a walk with no edge repeated. A circuit is a closed trail.

Definition 15.2.1.3: A path is a series of distinct vertices, i.e., a walk which never revisits any previous vertex and a cycle is a closed path.

Definition 15.2.1.4: A Hamilton cycle in a graph G is a cycle which visits every vertex of G . A graph which contains a Hamilton cycle is called Hamiltonian.

Lemma 15.2.1.1: Given vertices $u \neq w$ in a graph G , if there is a walk from u to w in G , then there is a path from u to w in G .

Proof: Let W be a walk in G consisting of the vertices v_i , $1 \leq i \leq k$, $v_1 = u$, $v_k = w$ and edges e_i , $1 \leq i \leq (k-1)$. If $v_1, \dots, v_k$ are all distinct, W is a path.

If not, let $x = v_j$ be the first occurrence in W of a vertex which appears more than once, and let $v_{j'}$ be the latest occurrence of that vertex in W . Let W' be the walk formed from W by removing the vertices $v_{j+1}, \dots, v_{j'}$ and the edges $e_j, \dots, e_{j'-1}$. Then x appears only once in W' and so the number of repeated vertices in W' is strictly smaller than W .

We can then repeat this process starting with the walk W' . As each time we repeat, the number of vertices appearing more than once strictly decreases, and eventually we must have a walk containing no repeated vertices, that is, a path.

□

Lemma 15.2.1.2: If every vertex in a finite graph G has degree > 1 , then G contains a cycle.

Proof: Let $|V| = n$. Starting from any vertex u in G , then as $d(u) > 1$, there is some edge $e = uv$ incident to u . Let $u = v_1$, $v = v_2$. We construct a cycle in G as follows: at v_2 , as $d(v_2) > 1$, there must be some edge $e_2 \neq e$ incident to v_2 . There are two possibilities:

- (a) There is an edge from v_i to some already visited v_k with $1 \leq k \leq i$.
- (b) We do not have (a), and there is some edge $e_i = \{v_i, v_{i+1}\}$ to some unvisited new vertex v_{i+1} .

But at some point $i \leq n$, we must have situation (a) occurring - we cannot keep finding new vertices. Then the path $v_k, v_{k+1}, \dots, v_i, v_k$ is a cycle in G as required.

□

15.2.2 Connectedness

Definition 15.2.2.1: A graph G is connected if, given any two vertices, there is a path between them.

Definition 15.2.2.2: A connected component of $G = (V, E)$ is a subgraph $G_1 = (V_1, E_1)$ of G such that:

- G_1 is connected.
- For any $u \in V_1$ and any $v \in V$ such that there is a path from u to v in G , then $v \in V_1$.
- For every edge $e = uv \in E$ where $u, v \in V_1$, then $e \in E_1$.

15.2.3 Euler's Theorem

Definition 15.2.3.1: An Euler circuit in a graph G is a circuit which uses each edge of G exactly once. A graph with an Euler circuit is called Eulerian.

Theorem 15.2.3.1: Let $G = (V, E)$ be a finite connected graph with $|V| > 1$. Then G has an Euler circuit $\iff$ all vertices of G have even degree.

Proof: We have already seen why $\implies$, to prove $\impliedby$, we require the following Lemma:

Lemma 15.2.3.1: If a graph $G = (V, E)$ is finite and connected, with $|V| > 1$ and all vertices of even degree, then G is a finite union of edge-disjoint cycles, that is, there are cycles $C_1, C_2, \dots, C_n$ such that the union of the vertices of these cycles is equal to V and the union of edges is equal to E .

Proof: Since G is connected, it has no isolated vertices, so all vertices have degree ≥ 2 , by Lemma 15.2.1.2, there is a cycle C_1 in G . We form a new graph from G by removing the all edges from C_1 . The vertices in C_1 have their degrees decreased by 2 and then discard any which have become isolated. We can then repeat the process until the whole graph has been decomposed into cycles.

□

Proof of Theorem 15.2.3.1 ($\impliedby$): Let G be finite, connected, and with all degrees even. Then, by Lemma 15.2.3.1, G is a finite union of n edge-disjoint cycles. We shall show, by induction on n , that these can be combined into a single Eulerian circuit.

- Base Case: If $n = 1$, the one cycle is the Euler circuit.
- Induction Hypothesis: If G is made up of $\geq (n - 1)$ edge-disjoint cycles, it has an Euler circuit.
- Inductive Step: Now consider a graph G made up of n edge-disjoint cycles,

$$G = C_1 \cup C_2 \cup \dots \cup C_{n-1} \cup C_n \quad (178)$$

And let

$$H = C_1 \cup C_2 \cup \dots \cup C_{n-1} \quad (179)$$

As in the proof of Lemma 15.2.3.1, H has all degrees even. It may not be connected, so consider its connected components, $H_1, H_2, \dots, H_k$. Since each cycle C_i lies within only one connected component H_j , we know that there are at most $(n - 1)$ connected components.

Furthermore, since the union $C_1 \cup C_2 \cup \dots \cup C_{n-1}$ makes up the whole of H , each H_j must be made up by $\leq (n - 1)$ edge-disjoint cycles. So, by the Induction Hypothesis, each connected component H_j has an Euler circuit E_j .

But since G was connected, C_n must visit each H_j , so we can make an Euler circuit for G by following C_n , but when we first visit each H_j , follow E_j all the way round before continuing.

□

15.3 Planar Graphs

15.3.1 The Three Utilities Problem

Definition 15.3.1.1: The three utilities problem asks whether we can connect gas, water, and electricity to three houses without the pipes/wires crossing?

Definition 15.3.1.2: Given a graph $G = (V, E)$ where there exists $V_1, V_2 \subseteq V$ such that $V_1 \cup V_2 = V$, $V_1 \cap V_2 = \emptyset$, and every edge of G is of the form uv where $u \in V_1$ and $v \in V_2$, then G is bipartite.

Definition 15.3.1.3: A graph G is planar if it can be drawn in the plane with no edges crossing. Such a drawing is a planar representation of G .

15.3.2 Faces and Boundaries

Definition 15.3.2.1: A planar representation of a graph G divides the plane into a set of F faces. The boundary of a face is the set of edges around it, considered as a walk in G . For a face $f \in F$, we write $b(f)$ for the length of the boundary of f .

Lemma 15.3.2.1: Planar Handshaking, for any planar representation of a graph $G = (V, E)$, we have

$$\sum_{f \in F} b(f) = 2|E| \quad (180)$$

Proof: Each edge contributes 2 to the sum of boundary lengths, one for each side of the edge.

□

Definition 15.3.2.2: A bridge in a graph G is an edge e of G such that removing e from G increases the number of connected components.

Lemma 15.3.2.2: Given any edge e of a graph G , then e is a bridge $\iff e$ is not part of a cycle in G .

Proof: Let e join the vertices u and v of G . We prove the contrapositive - if e is part of a cycle C , then there is a path P between u and v formed by removing e from C . Then e is not a bridge, since given any path between vertices w_1 and w_2 which passes through e , we can traverse P instead of the edge e to find a path between w_1 and w_2 .

As this is possible, removing e would not increase the number of connected components. Conversely, if e is not a bridge, then considering the graph $G' = G - e$, we have u, v are in the same connected component in G' , so there is a path from u to v in G' , and adding e to this path forms a cycle in G .

□

Theorem 15.3.2.1: The Jordan Curve Theorem states that, a simple closed curve in the plane divides it into two regions, inside and outside.

Proof: Proof omitted.

□

Theorem 15.3.2.2: In a planar representation of a finite graph, e is a bridge $\iff$ both sides of e are on the same face.

Proof: By Lemma 15.3.2.2, if e is not a bridge, it is contained in a cycle, which forms a simple closed curve in the plane, and so by the Jordan Curve Theorem the two sides of e are different regions.

□

15.3.3 Euler's Formula

Theorem 15.3.3.1: Euler's Formula, Let $G = (V, E)$ be a planar graph, with the set of faces F , and k connected components. Then

$$|V| - |E| + |F| = k + 1 \quad (181)$$

Proof: By induction on $|E|$.

- Base Case: When $|E| = 0$, G is a null graph, so $|F| = 1$ and $k = |V|$.
- Inductive Hypothesis: Assume Euler's formula is true for any graph with $|E| - 1$ edges.
- Inductive Step: Draw $G = (V, E)$, which has $|E|$ edges, in the plane. Remove one edge e to make G_0 . G_0 has $|V|$ vertices, $|E| - 1$ edges, and, say, $|F_0|$ faces and k_0 components. The induction hypothesis applies for G_0 , so we know that

$$|V| - (|E| - 1) + |F_0| = k_0 + 1 \quad (182)$$

$|F_0|$ and k_0 relate to $|F|$ and k in one of two cases, either:

1. e is a bridge in G . Removing it makes one more component in G_0 , so $k_0 = k + 1$. By Theorem 15.3.2.2, its two sides bound the same face, so $|F_0| = |F|$, and (182) becomes

$$|V| - (|E| - 1) + |F| = (k + 1) + 1 \quad (183)$$

2. e is not a bridge in G . Removing it does not disconnect G , so $k_0 = k$. But by Theorem 15.3.2.2, in G its sides bounded two distinct faces, which now become one. So (182) becomes

$$|V| - (|E| - 1) + (|F| - 1) = (k + 1) + 1 \quad (184)$$

In both cases, we have $|V| - |E| + |F| = k + 1$, so Euler's formula holds for G .

□

15.3.4 Simple Planar Graphs

Theorem 15.3.4.1: If $G = (V, E)$ is a simple planar graph with $|V| \geq 3$, then

$$|E| \leq 3|V| - 6 \quad (185)$$

Proof: Provided G has at least two edges, then every face has $b(f) \geq 3$, so Lemma 15.3.2.1 gives

$$2|E| = \sum_{f \in F} b(f) \geq 3|F| \quad (186)$$

But also,

$$|V| - |E| + |F| = k + 1 \quad (187)$$

So,

$$\begin{aligned} |V| - |E| + \frac{2}{3}|E| &= k + 1 \geq 2 \\ |V| - |E| + \frac{2}{3}|E| &\geq 2 \\ |V| - 2 &\geq \frac{|E|}{3} \end{aligned}$$

Which can be rearranged to give (185) as claimed.

□

Lemma 15.3.4.1: K_5 is not planar.

Proof: For $G = K_5$ we have that $|V| = 5$ and $|E| = 10$ so $|E| \not\leq 3|V| - 6$.

□

Definition 15.3.4.1: The girth g of G is the shortest length of a cycle in G . If G has no cycles, then $g = 0$.

Theorem 15.3.4.2: If G is simple, planar, and with girth $g \neq 0$, then

$$|E| \leq \frac{g}{g-2}(|V| - 2) \quad (188)$$

Proof: Here every face has $b(f) \geq g$, so we get

$$\frac{2|E|}{g} \geq |F| \quad (189)$$

from Lemma 15.3.2.1. Again, using $|V| - |E| + |F| = k + 1$, we get

$$|V| - |E| + \frac{(g-2)}{g}|E| \geq 2 \quad (190)$$

Which can be rearranged to get (188) as claimed.

□

Lemma 15.3.4.2: $K_{3,3}$ is not planar.

Proof: Since $K_{3,3}$ is bipartite, it has no cycles of odd length, and so we can see $g = 4$. We have $|V| = 6$ and $|E| = 9$, so

$$\frac{g}{g-2}(|V| - 2) = \frac{4}{2}(6 - 2) = 8 \quad (191)$$

and $|E| = 9 \not\leq 8$.

□

Definition 15.3.4.2: A subdivision of G is a graph obtained from G by subdividing edges of G , where subdividing an edge $e = uv$ if G means

- (a) Adding a new vertex w to G .
- (b) Adding edges uw, wv .
- (c) Removing the original edge e .

Theorem 15.3.4.3: From (Kuratowski, 1930), a graph G is planar $\iff G$ contains no subdivision of K_5 or $K_{3,3}$ as a subgraph.

Proof: $\Leftarrow$ follows from Lemma 15.3.4.1 and $\Rightarrow$ from Lemma 15.3.4.2.

□

Theorem 15.3.4.4: Every finite simple planar graph has a vertex of degree ≤ 5 .

Proof: Clearly true for $V \geq 6$. For $|V| > 6$, suppose all vertices are of degree ≥ 6 , then by handshaking, $|E| \geq 3|V|$. Then G does not satisfy Theorem 15.3.4.1 and so is not planar.

□

15.3.5 Platonic Graphs

Definition 15.3.5.1: A Platonic graph is finite, simple, planar, and connected, with all vertices of the same degree $d \geq 3$, and all faces of the same boundary length, $b \geq 3$.

d	b	 V 	 E 	F	
3	3	4	6	4	tetrahedron
3	4	8	12	6	cube (hexahedron)
4	3	6	12	8	octahedron
3	5	20	30	12	dodecahedron
5	3	12	30	20	icosahedron
from (194)	from (194)	from (193)	from (192)	from (192)	

Table 2: Platonic Graphs

Theorem 15.3.5.1: There are just five Platonic graphs.

Proof: By handshaking,

$$d|V| = 2|E| = b|F| \quad (192)$$

So Euler's formula $|V| - |E| + |F| = k + 1$ becomes

$$|V|(2b - bd + 2d) = 4b \quad (193)$$

But $|V| > 0$ and $b > 0$ so

$$(b - 2)(d - 2) < 4 \quad (194)$$

So we have Table 2.

□

Part VI

References

16 Algorithms and Data Structures

16.1 Module Professors

Fundamental Data Structures and *Graph Theory and Algorithms* are taught by Doctor David Kutner.

- **Office Location:** Room 2011, MCS Building
- **Website:** <https://www.durham.ac.uk/staff/david-c-kutner/>
- **Preferred Contact Methods:** david.c.kutner@durham.ac.uk
- **Pronouns:** he, him

Asymptotic Notation and Sorting is taught by Professor Thomas Erlebach.

- **Office Location:** Room 2004, MCS Building
- **Website:** <https://www.durham.ac.uk/staff/thomas-erlebach/>
- **Preferred Contact Methods:** thomas.erlebach@durham.ac.uk
- **Pronouns:** he, him

Selection and Data Structures is taught by Doctor Anish Jindal.

- **Office Location:** Room 2065, MCS Building
- **Website:** <https://www.durham.ac.uk/staff/anish-jindal/>
- **Preferred Contact Methods:** anish.jindal@durham.ac.uk
- **Pronouns:** he, him

16.2 Cited Works

Erickson, J. (2019). Algorithms. 1st ed. [online] Self-Published, p.71.
Available at: <http://jeffe.cs.illinois.edu/teaching/algorithms/> [Accessed 25 Mar. 2026].

Rosen, K.H. (2003). Instructor's Resource Guide for Discrete Mathematics and Its Applications.
Boston: McGraw-Hill, pp.634-636.

17 Computational Thinking

18 Computer Systems

18.1 Module Professors

Digital Electronics and *Machine Architecture* are taught by Doctor Ioannis Ivrisimtzis.

- **Office Location:** Room 2023, MCS Building
- **Preferred Contact Methods:** ioannis.ivrisimtzis@durham.ac.uk
- **Pronouns:** he, him

Databases is taught by Doctor Peter Davies-Peck.

- **Office Location:** Room 2063, MCS Building
- **Preferred Contact Methods:** peter.w.davies@durham.ac.uk
- **Pronouns:** he, him

Operating Systems is taught by Doctor Anish Jindal.

- **Office Location:** Room 2065, MCS Building
- **Preferred Contact Methods:** anish.jindal@durham.ac.uk
- **Pronouns:** he, him

18.2 Cited Works

n/a

19 Mathematics for Computer Systems

19.1 Module Professors

Calculus is taught by Doctor Junyan Hu and Doctor Sukanya Pandey.

Doctor Junyan Hu:

- **Office Location:** Room 2060, MCS Building
- **Preferred Contact Methods:** junyan.hu@durham.ac.uk
- **Pronouns:** he, him

Doctor Sukanya Pandey:

- **Office Location:** Room 2011, MCS Building
- **Preferred Contact Methods:** sukanya.pandey@durham.ac.uk
- **Pronouns:** he, him

Linear Algebra is taught jointly by Professor Steven Bradley and Doctor Karl Southern.

Professor Steven Bradley:

- **Office Location:** Room 2027, MCS Building
- **Preferred Contact Methods:** s.p.bradley@durham.ac.uk
- **Pronouns:** he, him

Doctor Karl Southern:

- **Office Location:** Room 2007, MCS Building
- **Preferred Contact Methods:** karl.southern@durham.ac.uk
- **Pronouns:** he, him

19.2 Cited Works

n/a

20 Discrete Mathematics

20.1 Module Professors

Introduction to Discrete Mathematics and *Combinatorics* was taught by Professor Mikhail Menshikov.

Graph Theory is taught by Irene Ayuso-Ventura.

- **Preferred Contact Methods:** irene.ayuso-ventura@durham.ac.uk

20.2 Cited Works

Perepelitsa, I. (2020). MATH 3336 - Discrete Mathematics. [online] University of Houston.
Available at: <https://www.math.uh.edu/irina/MATH3336/3336Notes/3336RR.pdf> [Accessed 16 May 2026].

Kuratowski, C. (1930). Sur Le Problème Des Courbes Gauches En Topologie. *Fundamenta Mathematicae*, 15, pp.271- 283. doi:10.4064/fm-15-1-271-283.